



NCAL Characterization

at NPI, ÚJV Řež

Working Document

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COLLABORATING INSTITUTIONS

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Chapter 1

Introduction

1.1 Physics and detector context

The Compressed Baryonic Matter (CBM) experiment at FAIR is designed to study strongly interacting matter at high net-baryon density under laboratory conditions [16]. Within that program, reliable event characterization is essential: centrality determination, reaction-plane reconstruction, and the interpretation of event-by-event observables all depend on a robust measurement of forward-going spectator matter. The forward detector system therefore has a direct impact on the quality of the physics extracted from heavy-ion data, especially in the high-rate environment foreseen for CBM.

The Forward Spectator Detector (FSD) has been developed as the dedicated forward detector for spectator nucleons and nuclear fragments in CBM. Its role is to provide event-plane information and support centrality reconstruction at the interaction rates relevant for the experiment [15]. However, charged spectators alone do not provide the full picture of the forward final state. A measurement of neutral forward particles, especially neutrons, is required if the forward system is to deliver a more complete description of spectator emission and its correlation with the collision geometry.

The Neutron Calorimeter (NCAL) addresses this missing neutral component. According to the current NCAL addendum, the detector is intended to complement the FSD by measuring forward neutral spectators and thereby strengthening centrality and collective-flow related measurements in the CBM heavy-ion program [21]. The same document also emphasizes a broader physics reach: forward neutron detection can support dedicated studies of elementary channels such as proton-proton and deuteron-proton reactions, and may provide a handle for initial-state neutron tagging in deuteron-induced interactions [21]. In that sense, NCAL is not merely an auxiliary subsystem, but a detector with its own performance questions and physics potential.

The present NCAL prototype program is built around repurposed detector components originating from the COSY Time-of-Flight system, combined into dedicated test configurations for commissioning and performance studies [21, 22]. Prototype setups have been used both in the mini-CBM environment at GSI and in external test campaigns, including measurements at NPI, ÚJV Řež[21]. These campaigns are scientifically important because they provide controlled conditions under which detector response can be studied before the final integration of NCAL into the CBM forward system.

1.2 Motivation for the present study

The main objective of the present work is to characterize the NCAL prototype at a level that is useful for detector development, future integration, and later publication. In practical terms, this means quantifying how the detector responds in a neutron-beam environment, identifying which observables carry the strongest discriminating power, and determining which aspects of the readout chain are stable enough to support reproducible analysis.

One central question is the neutron detection efficiency of the detector under the beam conditions available during the test campaign. That quantity is directly relevant for evaluating whether the prototype geometry, active material, and readout concept are adequate for the role envisioned for NCAL in the CBM forward system. Efficiency, however, is not the only quantity of interest. A useful neutron detector must also be understood in terms of timing behavior, prompt backgrounds, and event-selection quality. For that reason, the present study also treats particle-identification methods as a core topic rather than a secondary analysis exercise.

The detector-development aspect of this work is equally important. The campaign is not only a neutron-response measurement; it is also a test bench for readout and data-acquisition concepts. In particular, the study is intended to improve the understanding of NCAL performance when operated with the newly developed DOGMA front-end electronics and with the older DiRICH-based chain used in earlier detector work. Even when the final CBM readout choice lies outside the scope of this document, comparative operation with these chains is highly informative because it exposes differences in timing reference quality, signal handling, data integrity, and downstream analysis requirements. A careful comparison is therefore part of understanding the detector itself. The second campaign at NPI, ÚJV Řežshould therefore be understood primarily as a detector-characterization and readout-validation experiment rather than as a finalized production-data run.

1.2.1 Motivation from the first campaign at NPI, ÚJV Řež

The present campaign should be read explicitly as experiment 2. According to the current run history, the preceding campaign used a proton-on-lithium neutron-production target in a fixed-position configuration. In that first experiment, the gamma-induced population formed a broad structure directly beneath the neutron-candidate region in the time-of-flight versus pulse-height plane. Under those conditions, the overlap between the two populations made it very difficult to characterize the NCAL detector response cleanly.

Figure 1.1 documents that limitation using the time-of-flight-overlay version of the first-campaign plot. The figure is included here because it captures the central practical reason for carrying out a second campaign under modified conditions: the dominant lower population and the neutron-candidate accumulation were not sufficiently separated in the observable space available in experiment 1.

For comparison, the same figure also includes an analytical estimate of the prompt-gamma and 30 MeV neutron time-of-flight dependence on detector distance, together with the corresponding time-of-flight difference.

1.3 Scope of this document

This document is being developed as a broad scientific record of the second experimental campaign and is intended to mature into material suitable for a thesis chapter and, after further refinement, parts of one or more journal articles. Its purpose is therefore twofold. First, it should document the experiment rigorously enough that the measurement logic, detector configuration, and analysis choices remain transparent and reproducible. Second, it should separate verified results from working hypotheses, so that later interpretation can be strengthened without rewriting the experimental history.

The analysis workflow already points toward the structure needed for such a document. The existing event-digestion tools extract trigger-referenced observables such as leading and trailing edge times, time-over-threshold values, trigger-to-trigger spacing, and timing jitter from raw DOGMA-formatted data. These observables are natural building blocks for beam-structure studies, time-of-flight analysis, and future particle-identification strategies. Accordingly, the full document should evolve toward a coherent treatment of the experimental setup, beam conditions, detector geometry, veto configuration where applicable, DAQ architecture, calibration logic, event reconstruction, uncertainty sources, and performance observables.

At the present stage, the introduction is intentionally conservative. It establishes the detector and physics context from available project documents and defines the immediate scientific aims of this campaign: characterization of NCAL, evaluation of neutron-response observables, exploration of PID strategies, and assessment of detector operation together with DOGMA and DiRICH readout concepts. Subsequent chapters can build on this foundation by adding the finalized beamline description, analysis methodology, efficiency extraction strategy, uncertainty treatment, and validated experimental results.

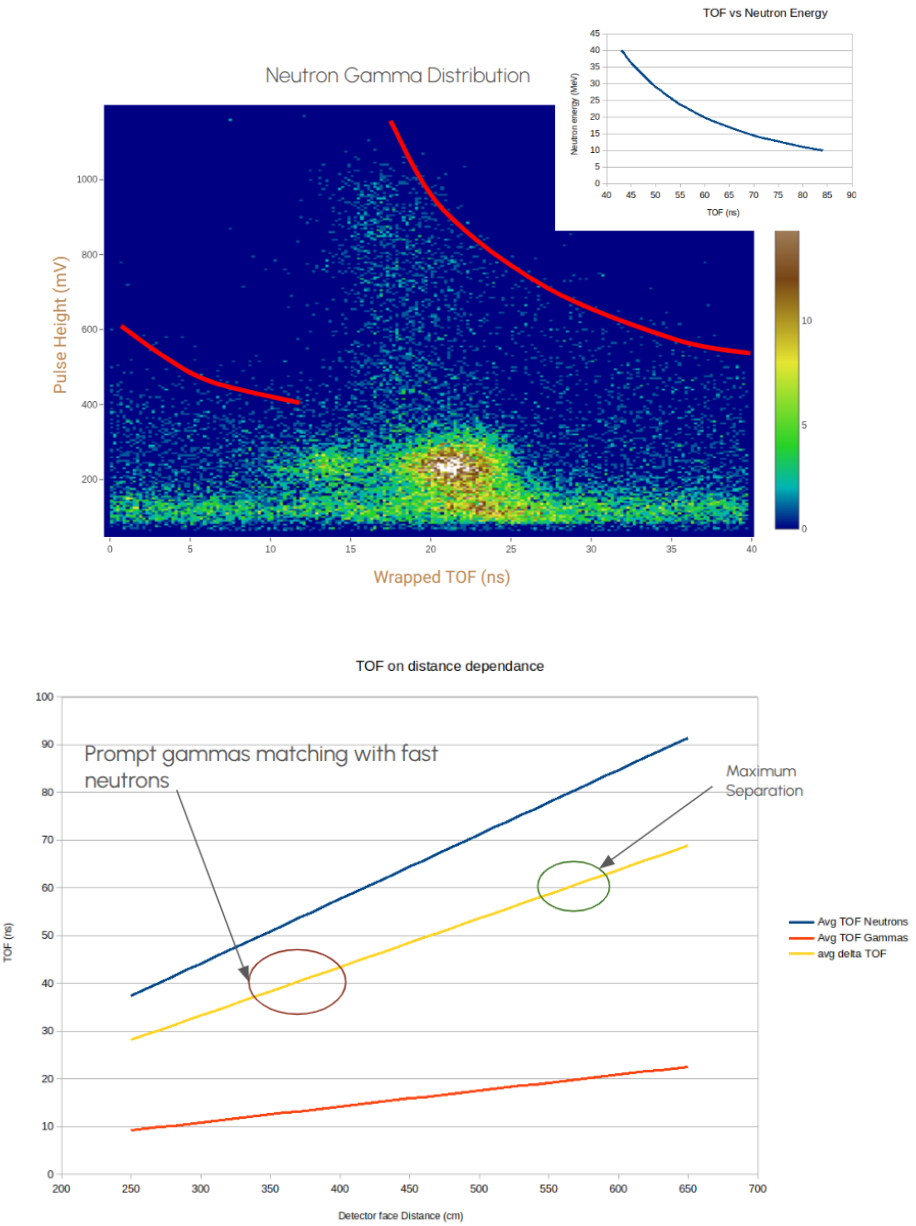


Figure 1.1: Top: wrapped time-of-flight versus pulse-height distribution from the first campaign at NPI, ÚJV Řež, shown in the time-of-flight-overlay form used during later interpretation of the dataset. In the working interpretation of experiment 1, the broad lower population corresponds to gamma-induced events, while the higher-time-of-flight structure contains the neutron-candidate region. Bottom: analytical estimate of prompt-gamma time of flight, 30 MeV neutron time of flight, and the corresponding time-of-flight difference as a function of detector-face distance. Together, the two panels illustrate why limited timing separation in the first campaign made a clean characterization of the NCAL response difficult.

Chapter 2

Experimental Setup

The second NCAL campaign at NPI, ÚJV Řežwas carried out in the proton-driven neutron environment of the U-120M cyclotron. This chapter summarizes the experimental arrangement relevant for interpreting detector response: the beamline geometry, detector placement, shielding layout, and the roles of the main supporting subsystems. Detailed source-reaction physics is treated in the next chapter, while detector construction and DAQ implementation are described later.

2.1 Setup geometry and detector placement

Figure 2.1 summarizes the layout adopted for experiment 2. The schematic records the current setup logic, detector ordering, and monitoring strategy along the beam axis, while also identifying those elements that remained under optimization during preparation of the campaign.

The experiment was performed in the forward-directed mixed neutron–gamma field produced by cyclotron protons on a beryllium target. The forward radiation field emerging from the target was sampled downstream along the axis of the planned NCAL position scan. The drawing presently labels the proton beam with an energy of 32.5 MeV and also marks the cyclotron RF structure as 25 MHz with a 40 ns period. These values are retained here as setup identifiers from the current experiment sketch and should be cross-checked against the run-time machine settings recorded during the actual campaign.

One distinctive feature of the planned geometry is the pair of mini scintillator plates placed immediately next to the target and oriented toward the production point. In the present concept they are intended to operate in coincidence mode and to provide a local indication that production is taking place near the target. Their role is explicitly secondary: they are currently treated as a backup or plan-B monitoring option rather than as a validated primary beam monitor. Their efficiency in the actual source geometry, their count-rate capability, and even their practical usefulness under the expected radiation conditions are not yet established and will need to be tested experimentally.

Farther downstream, the first gamma detector is intended to monitor the prompt and background gamma field before the main shielding block. The present expectation is that this detector may provide both gamma-spectral information and useful timing information associated with prompt gamma production. A 1.2 m thick shielding block with an 8×8 cm^2 aperture is then used to define the line of sight toward the central NCAL module. In the current layout logic, the aperture is not merely a mechanical opening but part of the background-reduction strategy: it restricts the

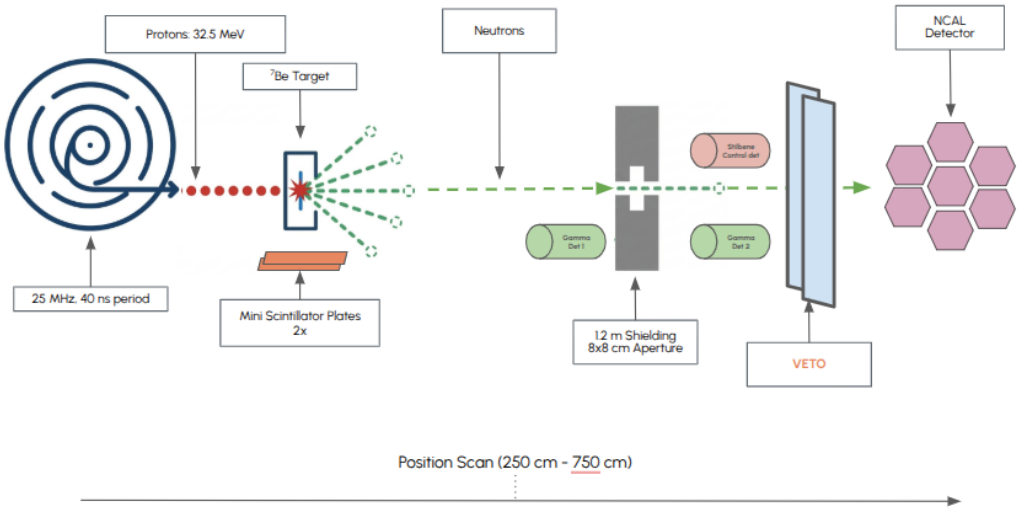


Figure 2.1: Schematic layout of the second experiment at NPI, ÚJV Řež. The scheme shows the proton beam delivered from the cyclotron onto a beryllium target, target-adjacent mini scintillator plates planned as a coincidence-based backup production monitor, an upstream gamma detector, a $1.2\,\mathrm{m}$ shielding block with an $8 \times 8\ \mathrm{cm}^2$ aperture aligned to the central NCAL line of sight, a downstream gamma detector behind the shield, a compact control stilbene detector near the NCAL entrance plane, the seven-module NCAL flower geometry, and an optional thin plastic veto layer in front of the central module. The position-scan direction indicated in the sketch corresponds to the current working range of roughly $250\,\mathrm{cm}$ to $750\,\mathrm{cm}$. Several elements, especially the target-adjacent scintillators, the exact control-detector placement, and the use of the front veto layer, remained part of the ongoing setup optimization.

direct acceptance to the intended central region while allowing the surrounding shield to suppress off-axis radiation.

Behind this shielding block, a second gamma detector is planned upstream of NCAL but in a substantially more shielded location than the first one. Its intended role is to characterize the radiation field that survives the shielding and reaches the NCAL entrance region. In the most conservative wording, this means measuring the gamma spectrum behind the shielding; depending on detector response and local mixed-field conditions, the same detector may also record neutron-induced contributions, but that possibility should be treated as an effect to be measured rather than assumed a priori.

The control stilbene detector is presently envisaged near the NCAL entrance plane as a reference neutron monitor with known neutron-detection performance. Its exact position, however, remains an open optimization problem. From one point of view, the most direct comparison would place it on the same central line of sight as the central NCAL module. From another point of view, material placed directly in front of NCAL may partially shadow the calorimeter acceptance or generate additional secondary particles. For that reason, the final measurement sequence may need to alternate between different control-detector positions, or between control-detector and unobstructed NCAL configurations, instead of assuming a single permanent placement from the outset.

The NCAL detector itself is planned in the seven-module flower arrangement consisting of one central module surrounded by six peripheral modules in a honeycomb-like hexagonal pattern. The central module is aligned with the shielding aperture and therefore defines the primary line of interest in the present setup concept. In addition, a thin plastic scintillator layer may be placed in front of the central module as a veto for charged particles. At present this veto layer should be regarded as a diagnostic option rather than an established component of the finalized setup. The current expectation from the local beamline side is that charged particles should not emerge from the target region in a way that significantly reaches the NCAL position, but this statement still requires direct confirmation under the actual experimental conditions.

2.2 Experimental subsystems and their roles

Although the NCAL prototype is the detector under study, the full experiment involves several coupled subsystems. Their roles are summarized in Table 2.1. This distinction is important because later chapters discuss some of these elements separately in terms of hardware documentation, but at experiment level they function together as one measurement system.

The upstream and downstream gamma detectors, the control stilbene detector, and the target-adjacent mini scintillators each probe a different part of the radiation field and together provide the contextual information needed to interpret the NCAL response. The shielding and aperture assembly likewise form an active part of the measurement definition, because they determine which fraction of the field is intentionally delivered to the calorimeter entrance.

Table 2.1: Main experimental subsystems and their role in the present campaign.

Subsystem	Role in the experiment	Main observables or function
NCAL prototype	Detector under study	Neutron response, timing behavior, detector stability, and future efficiency-related observables [21, 22].
Target-adjacent mini scintillator pair (planned)	Backup production monitor near the target	Local coincidence signal near the production point; presently a contingency monitor whose practical efficiency is not yet established.
First gamma detector	Upstream mixed-field monitor	Prompt-gamma timing information and gamma-spectrum monitoring before the main shielding block.
Shielding block with central aperture	Passive background reduction and geometrical selection	Restricts the direct line of sight toward the central NCAL module while suppressing off-axis radiation.
Second gamma detector	Downstream radiation-field monitor	Characterization of the radiation field behind shielding and in front of NCAL, primarily gamma-sensitive but potentially recording mixed-field contributions.
Control stilbene detector	Reference neutron monitor	Relative neutron-flux monitoring close to the NCAL entrance region, with placement still under optimization.
Optional thin plastic veto layer	Charged-particle check at the NCAL entrance plane	Diagnostic veto option intended to test whether any charged-particle contamination reaches the calorimeter position.
DAQ and front-end chains	Signal transport and event building	Trigger-referenced observables such as timestamps, time-over-threshold values, timing jitter, waveform shapes, and firmware-derived quantities.

2.3 Measurement strategy

The measurement strategy is centered on the NCAL flower geometry aligned to the shielding aperture and studied along the beam axis over the planned source-to-detector distance range indicated in Fig. 2.1. Supporting detectors are used to characterize different components of the local radiation field: the upstream gamma detector samples the prompt field before shielding, the downstream gamma detector characterizes the radiation that reaches the NCAL entrance region, and the control stilbene detector provides a neutron-sensitive reference close to the calorimeter position.

Additional configuration changes remain part of the experimental strategy where required by the measurement conditions. In particular, the exact placement of the control stilbene detector may be alternated to avoid permanent shadowing of the central NCAL line of sight, and the thin plastic veto layer may be inserted as a diagnostic configuration to test for any charged-particle contribution at the calorimeter entrance. Within this framework, the later analysis can focus on direct NCAL observables together with the contextual timing and background information delivered by the supporting detector chains.

Chapter 3

Production Physics

3.1 Beryllium-target neutron production at NPI

For the present NCAL campaign, the relevant source environment is the proton-driven neutron production line operated at the U-120M cyclotron at NPI Řež. The source literature available in this workspace shows that two related, but physically distinct, beryllium-based configurations are used there and should not be conflated. In the thick-target configuration, 35 MeV protons are fully stopped in an 8 mm beryllium target and produce a continuous fast-neutron spectrum extending up to about 33 MeV [20]. In the collimated-beam configuration, protons interact with a 2.5 mm beryllium foil, lose only part of their energy in the beryllium, and are then stopped downstream in a carbon slab; in that case the forward spectrum retains a quasi-monoenergetic character [2].

This distinction matters for any discussion of the accompanying gamma field. A statement such as "gamma spectrum for 35 MeV protons on thick Be" is incomplete unless the surrounding station geometry is also specified. The gamma background depends not only on the primary proton–beryllium interaction, but also on which additional beamline elements are struck by the proton beam and which materials are present close to the source.

The neutron-production papers also show that some structures in the forward spectrum arise from different $^{9}\mathrm{Be}(p, n)$ channels feeding the ground and low-lying excited states of ^{9}B [2, 20]. These structures are neutron-spectrum features. They should not be reinterpreted as evidence for prominent prompt gamma lines from ^{9}B* de-excitation.

3.2 Dominant neutron-production channels

The reviewed NPI source papers support a compact interpretation of the forward neutron spectrum at proton energies around 35 MeV. Thin-target measurements cited in these works show two high-energy components associated with the $^{9}\mathrm{Be}(p, n)$ charge-exchange reaction, corresponding to population of the ground state of ^{9}B and of its first two excited states at 2345 keV and 2751 keV, superimposed on a broader continuum from compound-nucleus reactions [2, 20]. In the thick-target configuration, the measured forward spectrum is then interpreted as the superposition of such partial spectra as the incident protons slow down in the beryllium. This produces the characteristic step-like structure observed near 30 MeV to 32 MeV [20].

Consequently, a 30 MeV neutron used later for analytical time-of-flight reference plots should be

read as a representative fast-neutron scale within the high-energy part of the forward spectrum, not as a statement that the source is monoenergetic at 30 MeV. The thick-target reference spectrum extends to about 33 MeV, while the prominent high-energy structure lies near 30 MeV to 32 MeV; lower-energy continuum neutrons are also present and will arrive later than the 30 MeV reference curve.

Table 3.1: Source-supported neutron-production channels relevant for the forward spectrum from 35 MeV protons on a thick beryllium target.

Channel	Spectral manifestation	Comment
$^{9}\mathrm{Be}(p,n)^{9}\mathrm{B}_{\mathrm{g.s.}}$	Highest-energy component, extending to about 33 MeV	Forms the upper part of the high-energy peak structure in the forward spectrum [20].
$^{9}\mathrm{Be}(p,n)^{9}\mathrm{B}^{*}$ to the first two excited states at 2345 keV and 2751 keV	Second component about 2.5 MeV lower than the highest peak	Together with the ground-state channel it produces the step-like structure near 30 MeV to 32 MeV. The literature cited by Majerle et al. indicates an approximate 6:4 area ratio between the upper and lower high-energy components [20].
Compound-nucleus contribution	Broad continuum underlying the peaks and extending to lower neutron energies	Described explicitly in the NPI beam papers as the continuum component of the spectrum [2, 20].

The same thick-target study also shows that very close to the source the measured spectrum is modified by scattering in the target-station construction materials and by the finite extent of the production region. For that specific NPI configuration, Monte Carlo simulations indicate that this effect becomes negligible by about 50 cm from the beryllium slab, after which inverse-square scaling is used for the flux normalization [20]. For scan distances of order 2.5 m to 7.5 m, this supports the use of $1/r^2$ as a first-order geometric scaling law, provided the same source-station configuration is maintained.

3.3 Expected gamma contributions

Only a limited set of gamma contributions can be stated with confidence from the presently available beamline literature. Since the gamma field depends on the exact source-station geometry, beam centering, and the distribution of carbon- and hydrogen-containing materials near the target, a detailed line-by-line intensity model is not introduced here. Instead, only source terms directly

supported by the reviewed NPI references are summarized.

The best established prompt contribution is the 4.439 MeV line from excited ^{12}C. In the collimated-beam study, the prompt gamma timing peak is attributed to proton-beam interaction with the carbon stopping slab and the subsequent de-excitation of ^{12}C, explicitly identifying the 4439 keV transition [2]. In the thick-target neutron-generator study, prompt gamma peaks are likewise associated with proton-beam interaction with a carbon beam collimator located upstream of the beryllium target [20]. A prompt carbon line near 4.44 MeV is therefore a realistic expectation whenever carbon components in the source station intercept part of the proton beam.

The same collimated-beam study also discusses the 2224 keV gamma from the $^{1}\mathrm{H}(n,\gamma)$ reaction [2]. This line is relevant for the local radiation environment and detector response, but it should be interpreted as a secondary environmental gamma from neutron capture in hydrogenous material rather than as a primary prompt component of proton–beryllium production.

The available references do not justify fixed yield fractions for individual gamma lines, and they do not provide sufficient support for a detailed source decomposition involving additional discrete lines or a dominant continuum component for the present geometry. Those features should therefore be introduced only after dedicated gamma spectroscopy or transport calculations for the finalized setup.

Table 3.2: Working expectations for gamma contributions relevant to beryllium-target operation at NPI Řež.

Source region	Signature	Comment
Carbon beamline components	Prompt line near 4.439 MeV	Supported when the proton beam intercepts carbon elements such as the beam collimator or carbon stopping slab [2, 20].
Hydrogenous material near the source or detector	Secondary line at 2.224 MeV	Associated with $^{1}\mathrm{H}(n,\gamma)^{2}\mathrm{H}$; relevant for environmental background and calibration-related response [2].
Other prompt components	Geometry dependent	Additional lines or continuum components should not be quantified without a run-specific measurement or transport simulation.

3.4 Implications for the NCAL analysis

For the present NCAL document, the safest working assumption is therefore a limited one: the prompt-gamma field should be expected to contain at least a strong timing component associated with fast source gammas, and a 4.439 MeV carbon line is physically plausible whenever carbon beamline components are irradiated. Beyond that, the detailed line content and relative intensities

should be considered geometry-dependent and not fixed a priori. In particular, if the actual run conditions differ from the 35 MeV reference papers in proton energy, beam centering, source hardware, shielding composition, or aperture arrangement, then the gamma spectrum must be re-evaluated for the exact run geometry rather than copied from a generic source table.

Accordingly, the present document treats the gamma spectrum only at the level justified by currently available evidence. Any quantitative spectral decomposition should be based on either a dedicated gamma measurement with an energy-calibrated detector or a transport calculation using the finalized source-station geometry and realistic material composition. Until such a study is completed, the table above should be regarded as a compact working expectation rather than a final spectral model.

Chapter 4

Experimental Detector Systems

This chapter summarizes the detector systems currently used in the experiment and the parts of their readout chains that matter for operation and data interpretation. The present hardware inventory includes a primary gamma-spectroscopy detector, consisting of an ORTEC 905-4 NaI(Tl) detector head coupled to an ORTEC Model 276 photomultiplier base with integrated preamplifier, two auxiliary thin scintillation counters instrumented with compact Hamamatsu PMT hardware, and one compact stilbene-based parallel monitor detector provided by the beamline team at NPI, ÚJV Řež. The discussion is organized by detector system rather than by vendor component alone, because in practical operation the experimentally relevant object is the full detector-plus-readout chain.

4.1 Detector inventory and intended role

Table 4.1 summarizes the detector systems presently identified for the campaign in a compact form suitable for later comparison of detector medium, sensor chain, biasing, signal compatibility, and experimental role. Where a parameter has not yet been verified directly in the available documentation or campaign notes, it is marked explicitly as not yet verified rather than inferred.

Table 4.1: Summary of detector systems presently identified for the campaign.

Detector	Detector medium	Sensor / Readout	Qty	HV range	Signal range	Signal polarity	Role
ORTEC 905-4 chain	3-inch $\times$ 3-inch NaI(Tl) crystal	3-inch PMT + ORTEC 276 base/preamplifier	1	Positive HV up to $2000\,\mathrm{V}$ at the base	Branch dependent; preamp and anode outputs serve different purposes	Positive preamp output; negative anode output	Gamma spectroscopy; calibration; threshold studies; commissioning support
Thin plastic scintillation counters	Thin plastic detectors; dimensions N/A	Hamamatsu E1761-04 socket assembly for 10 mm PMTs	2	Negative HV up to $1500\,\mathrm{V}$	Setup-specific pulse amplitude not yet verified in archived notes	Negative	Charged-particle or prompt-timing support; coincidence studies; auxiliary trigger work
SCIONIX stilbene monitor	$5 \times 5 \times 15$ mm crystalline stilbene	Hamamatsu R12421 PMT, direct raw-anode readout	1	Negative HV in assembled probe; PMT $\lvert V_{\mathrm{AK}}\rvert$ up to $1250\,\mathrm{V}$	Several-hundred-ns negative anode pulses; typically few-hundred-mV into $50\,\Omega$, possibly approaching about $1\,\mathrm{V}$ in current working notes	Negative	Parallel neutron monitor and relative beam-flux reference

Sources: Gamma-detector entry compiled from the ORTEC 905-series detector documentation and the ORTEC Model 276 manual [1, 23]. Thin-plastic-detector readout and high-voltage limits from the Hamamatsu E1761-04 documentation [18]. Stilbene-monitor PMT and high-voltage limits from the Hamamatsu R12421 datasheet [17]; assembly identification and setup-specific signal working values from current campaign notes.

4.2 Primary gamma-spectroscopy detector chain

The primary detector chain for the auxiliary gamma-ray measurements and calibration studies is a conventional scintillation assembly composed of an ORTEC 905-4 NaI(Tl) detector head and an ORTEC Model 276 photomultiplier base with integrated preamplifier. In the local hardware inventory, the detector head is additionally identified as an Alpha Spectra, Inc. model 12I12/3. These two components should be treated as a single measurement chain: the NaI(Tl) crystal converts deposited gamma-ray energy into scintillation light, the photomultiplier tube (PMT) converts that light into charge, and the Model 276 base supplies both the dynode bias distribution and the first analog conditioning stage [1, 23].

4.2.1 ORTEC 905-4 NaI(Tl) detector head

The ORTEC 905 series is a family of NaI(Tl) scintillation detectors intended primarily for gamma-ray counting and spectroscopy [1]. For the present work, the relevant unit is the 905-4 configuration, which combines a 3-inch diameter by 3-inch long NaI(Tl) crystal with a 3-inch PMT. This geometry is a standard compromise between absolute efficiency, moderate intrinsic energy resolution, and mechanical simplicity. Because iodine has a relatively high atomic number, NaI(Tl) retains a comparatively large full-energy absorption probability for laboratory gamma sources, which is one reason why this detector class remains a practical choice for commissioning, calibration, and source-based verification.

The 905-series datasheet emphasizes the standard physical advantages and limitations of NaI(Tl) detectors [1]. Their light output is sufficiently large for routine spectroscopy, and the manufacturer quotes a typical resolution of about 7% at ^{137}Cs for a 3-inch by 3-inch crystal, with somewhat worse values for larger detectors. At the same time, NaI(Tl) is not a particularly fast scintillator: the light decay constant is about $0.23\,\mu$s, and with a typical charge-sensitive front end this corresponds to an output-voltage rise time of about $0.5\,\mu$s. Consequently, the detector is well suited to efficiency-oriented gamma counting and moderate-resolution energy measurements, but it is not the natural choice when the best possible sub-nanosecond timing performance is required.

For the present report, two practical remarks from the manufacturer documentation are especially relevant. First, the 905-series detector heads are supplied to ORTEC by specialized detector manufacturers, so minor parameter differences between nominally identical detector heads are possible [1]. Second, the datasheet quotes representative total counting efficiencies for a 1-μCi ^{137}Cs source positioned at a distance of $10\,$cm; these values are useful as order-of-magnitude expectations during laboratory checkout, but they should not be mistaken for a substitute for an experiment-specific efficiency calibration.

Table 4.2: Manufacturer information most relevant to the ORTEC 905-4 detector used in the present setup.

Parameter	Value	Comment
Detector family	ORTEC 905 series NaI(Tl) scintillation detector	Modular scintillation-detector family for gamma-ray measurements [1].
Nominal detector size	3-inch diameter $\times$ 3-inch length crystal	ORTEC model 905-4; the local assembly is identified as Alpha Spectra model 12112/3.
Photomultiplier size	3-inch PMT	Integral part of the detector head supplied with the 905-4 configuration [1].
Typical energy resolution	About 7% at ^{137}Cs for a 3-inch by 3-inch crystal	Representative manufacturer value for routine gamma spectroscopy.
Light decay constant	About $0.23\,\mu$s	Sets a lower bound on the detector pulse width and therefore influences timing performance.
Typical voltage-pulse rise time	About $0.5\,\mu$s with a typical charge-sensitive preamplifier	Useful for estimating shaping or record-length requirements in downstream electronics.
Typical total counting efficiency	2.00% at 0.5 MeV and 1.30% at 2.0 MeV for a 1-μCi ^{137}Cs source at $10\,$cm	Quoted specifically for the 905-4 geometry; intended as a comparative detector figure rather than an in-beam calibration.

4.2.2 ORTEC Model 276 photomultiplier base with integrated preamplifier

The ORTEC Model 276 is not merely a passive PMT socket. It is a combined PMT base and analog front-end designed for essentially all 10-stage photomultipliers that fit the standard 14-pin socket geometry [23]. The instrument contains a resistor-divider bleeder chain for dynode biasing, a focus adjustment for optimizing the photocathode-to-first-dynode field, an ac-coupled anode output intended mainly for timing applications, and a self-contained low-noise integrating preamplifier coupled to the last dynode for spectroscopy-oriented energy readout [23]. In practical terms, the device therefore separates the detector signal into two complementary observables: a fast negative anode pulse for timing work and a slower positive preamplifier pulse for energy analysis.

The manufacturer specifications show why the Model 276 is suitable for routine scintillation spectroscopy. The PMT divider has a total resistance of $1.49\,\mathrm{M}\Omega$ and draws about $1.33\,\mathrm{mA}$ at the maximum applied voltage of $2000\,\mathrm{V}$. The integrated preamplifier has a nominal conversion gain of $5\,\mu\mathrm{V\,eV}^{-1}$ for a reference configuration consisting of a 2-inch by 2-inch NaI(Tl) crystal and a PMT gain of 10^6; the same manual quotes a rise time below $100\,\mathrm{ns}$ for a fast test pulse, a fall-time constant of $50\,\mu\mathrm{s}$, and an output noise below $50\,\mu\mathrm{V}$ rms [23]. These numbers should be interpreted as front-end performance parameters rather than as immutable detector properties, because the actual pulse amplitude still depends on the detector head, the PMT installed, and the chosen high-voltage operating point.

The Model 276 manual also provides the key operational constraints relevant to the present work. The PMT base expects a positive high-voltage supply with a maximum value of $2000\,\mathrm{V}$, the detector–PMT assembly must be made fully light-tight, and the single-turn focus potentiometer must be adjusted under irradiation so that the pulse amplitude is maximized and then mechanically locked [23]. The manual further notes that the photomultiplier gain changes by approximately a factor of two for each $100\,\mathrm{V}$ change in high voltage, so the actual gain setting is highly sensitive to the chosen bias voltage.

4.2.3 Operational settings relevant to the ORTEC NaI(Tl) chain

For the present NCAL-related laboratory work, the most important result of the manufacturer documentation is that there is no single universal bias setting that can be adopted blindly. The recommended procedure is instead operational: the detector must first be made light-tight, a positive high-voltage supply is then connected, and the high voltage is increased slowly while the preamplifier output is observed in the presence of a radiation source [23]. If a large amount of unipolar "grass" appears on the oscilloscope trace, the manual recommends checking explicitly for a light leak by covering the PMT region with an opaque cloth. Only after the detector has been verified to be optically sealed should the operating voltage be optimized.

The focus control is the second essential setting. According to the Model 276 manual, the correct procedure is to set the high voltage near the desired operating level, observe radiation-induced

Table 4.3: Selected specifications of the ORTEC Model 276 relevant to detector operation and DAQ coupling.

Parameter	Value	Comment
PMT compatibility	Standard 14-pin, 10-stage photomultipliers	Designed to provide near-optimum voltage distribution for this PMT class [23].
Maximum high voltage	Positive $2000\,\mathrm{V}$	The supply polarity is positive, with the cathode at ground potential.
Divider resistance	$1.49\,\mathrm{M\Omega}$ total bleeder resistance	Corresponds to a bleeder current of about $1.33\,\mathrm{mA}$ at $2000\,\mathrm{V}$.
Focus adjustment	External single-turn locking potentiometer	Used to optimize the photocathode-to-first-dynode field under operating conditions.
Preamplifier output	Positive, dc-coupled energy-analysis signal	Intended primarily for spectroscopy-oriented acquisition.
Anode output	Negative, ac-coupled pulse	Intended primarily for timing and coincidence work.
Nominal preamplifier conversion gain	$5\,\mu\mathrm{V\,eV^{-1}}$ in the reference 2-inch by 2-inch NaI(Tl) configuration	Reference value only; the actual output depends on detector, PMT, and high voltage.
Preamplifier rise time	$< 100\,\mathrm{ns}$ for a fast test pulse	Indicates that the base itself does not dominate the slower NaI(Tl) scintillation time scale.
Preamplifier fall-time constant	$50\,\mu\mathrm{s}$	Relevant for shaping and pile-up considerations at higher count rates.
Output noise	$< 50\,\mu\mathrm{V}$ rms	Useful lower bound when evaluating baseline quality.
Power requirements	$+24\,\mathrm{V}$ at $16\,\mathrm{mA}$ and $-24\,\mathrm{V}$ at $16\,\mathrm{mA}$	Compatible with ORTEC main amplifiers or the ORTEC 114 preamplifier power supply [23].

pulses, adjust the focus for maximum pulse amplitude, and then lock the potentiometer mechanically [23]. This is a physically meaningful adjustment rather than a cosmetic one, because it optimizes electron collection from the photocathode to the first dynode and therefore influences both gain stability and pulse-height response.

For routine acquisition, the manufacturer explicitly advises using the minimum practical high voltage when high count rates are expected, because count-rate tolerance is directly connected to the PMT gain and the gain changes by roughly a factor of two per $100\,\mathrm{V}$ [23]. In the context of a digitizer-based DAQ chain, this has two immediate consequences. First, the applied bias voltage should be documented together with each accepted configuration, because modest voltage changes can shift the effective gain substantially. Second, the digitizer settings must match the signal branch that is actually digitized: the preamplifier output is the natural branch for energy analysis and has positive polarity, whereas the anode output is the natural branch for timing studies and has negative polarity. A threshold or polarity configuration validated on one branch should therefore not be transferred to the other without re-optimization.

The manual also offers a simple functional calibration check that is useful during bench setup: a $1\,\mathrm{V}$ test pulse injected into the TEST input should yield a $1\,\mathrm{V}$ pulse at the preamplifier output [23]. This provides a straightforward verification that the base, preamplifier power, and downstream readout chain are behaving consistently before radioactive-source measurements are attempted.

Taken together, the ORTEC 905-4 and the ORTEC 276 form a robust, conventional scintillation-detector chain for laboratory gamma spectroscopy and detector commissioning. The detector head offers the expected efficiency advantages of a 3-inch by 3-inch NaI(Tl) crystal, while the base and preamplifier provide the biasing, gain control, and signal extraction needed to couple the assembly to either classical NIM electronics or a modern waveform digitizer. For the present campaign, these characteristics make the system well suited to calibration runs, threshold studies, and first-response checks of the broader NCAL readout infrastructure.

4.3 Auxiliary thin plastic detectors

The setup also includes two thin plastic detectors used as auxiliary counters. The presently available hardware identification for their optical readout is the Hamamatsu E1761-04 D-type socket assembly, which Hamamatsu specifies for 10 mm diameter head-on photomultiplier tubes [18]. This places the readout in the class of compact PMT assemblies rather than large-area spectroscopy bases. In practical detector terms, that geometry is well matched to lightweight auxiliary detectors, where compactness and prompt signal pickup are more important than the absolute full-energy efficiency expected from a much larger NaI(Tl) head.

The manufacturer specification most relevant to detector operation is electrical rather than mechanical. Hamamatsu lists the E1761-04 with negative high-voltage polarity, a maximum supply voltage of $1500\,\mathrm{V}$, and the same $1500\,\mathrm{V}$ maximum rating between the case and connector pins

[18]. The integrated voltage-divider chain has a total resistance of $3.63\,\mathrm{M}\Omega$, corresponding to a maximum divider current of 0.41 mA at the rated voltage. The product page further quotes a maximum leakage current of 1.0×10^{-10} A at the signal output and a maximum linear dc output current of $20\,\mu$A at 1500 V [18]. Current setup notes additionally identify the readout polarity used for these detectors as negative. Taken together, these are the main verified limits and operating characteristics that matter when the counters are connected to external high voltage, discriminators, or digitizer inputs.

At present, the local notes establish the detector bodies as thin plastic detectors mounted with Hamamatsu E1761-04 assemblies. The exact scintillator trade name, active area, and thickness are not yet verified from manufacturer documentation available in the workspace. Rather than introducing unsupported dimensions or assigning a plastic formulation that is not documented locally, Table 4.4 records the verified detector-type and readout specifications and leaves the remaining body parameters to be added later, once the counters are measured directly or their original labels are recovered.

4.4 Facility-provided parallel stilbene monitor

In addition to the detector systems brought by the NCAL group, the beamline team at NPI, ÚJV Řežindicated that one compact stilbene-based monitor detector would be operated in parallel with the NCAL measurements as an online flux reference. In the available campaign coordination notes, this detector is identified as a SCIONIX model `V5B15/0.5-E1-STILB-X-NEG`, consisting of a $5 \times 5 \times 15$ mm stilbene crystal optically coupled to a Hamamatsu `R12421` photomultiplier tube. A dedicated public datasheet for this exact assembled probe is not presently archived in the workspace. The description below therefore separates three layers of information that should not be conflated: assembly-level identification from campaign notes, material-level properties of crystalline stilbene from vendor documentation, and tube-level characteristics of the Hamamatsu photomultiplier from its manufacturer datasheet.

As a detector medium, stilbene belongs to the class of hydrogen-rich organic scintillators used for fast-neutron detection. Scionix summarizes the basic detection logic by noting that fast neutrons are detected in organic scintillators mainly through recoil protons, and that neutron–gamma separation is possible by pulse-shape discrimination when neutron and gamma interactions lead to different scintillation decay behavior [24]. The Inrad application note describes crystalline stilbene as particularly suitable for fast-neutron identification in the presence of gamma background and lists a density of $1.15\,\mathrm{g\,cm^{-3}}$, a maximum emission wavelength of about 380 nm, a prompt decay time of about 4.3 ns, and a rise time of about 0.1 ns [19]. These are material-level quantities rather than guaranteed performance numbers for the specific SCIONIX assembly used here, but they explain why stilbene is a scientifically reasonable choice for a compact reference detector in a mixed neutron–gamma field.

Table 4.4: Verified readout specifications for each of the two auxiliary thin scintillation counters equipped with Hamamatsu E1761-04 hardware.

Parameter	Value	Comment
Number of counters in setup	2	Pair of auxiliary thin scintillation counters planned for the present campaign.
Detector-body type	Thin plastic detector	Established in current setup notes; exact plastic grade and dimensions remain unverified.
Readout assembly	Hamamatsu E1761-04 D-type socket assembly	Manufacturer-designated PMT socket assembly used on the counters [18].
Applicable PMT	10 mm head-on type	Hamamatsu specification for the E1761-04 assembly.
Readout signal polarity	Negative	Current setup notes identify negative signal polarity for these counters.
Supply voltage polarity	Negative	The assembly is intended for negative high-voltage operation.
Maximum supply voltage	$1500\,\mathrm{V}$	Manufacturer maximum rating for operation.
Maximum insulation voltage (case to pins)	$1500\,\mathrm{V}$	Important when integrating the assembly into grounded mechanical supports.
Total voltage-divider resistance	$3.63\,\mathrm{M\Omega}$	Sets the divider current and power dissipation at the chosen operating voltage.
Maximum voltage-divider current	$0.41\,\mathrm{mA}$	Quoted at rated voltage by Hamamatsu.
Maximum leakage current in signal output	$1.0 \times 10^{-10}\,\mathrm{A}$	Upper bound from the product specification.
Maximum linear dc output current	$20\,\mu\mathrm{A}$ at $1500\,\mathrm{V}$	Useful when estimating acceptable load conditions.
Exact plastic grade / dimensions presently verified	Not yet identified	Active area, thickness, and scintillator trade name should be added after direct inspection of the two counters.

The same stilbene note describes neutron–gamma separation in terms of different prompt and delayed fluorescence fractions and presents vendor data with a pulse-shape-discrimination figure of merit of 4.7 near 500 keVee, with separation demonstrated down to 22 keVee and reported gamma misclassification rates of 10^{-6} [19]. For the present report, however, those figures should be treated only as indicative literature-level performance for high-quality stilbene and an optimized PSD chain. They should not be promoted to established performance of the particular 5 $\times$ 5 $\times$ 15 mm control detector unless the assembled probe is characterized directly under the same acquisition conditions used in the campaign.

At the photosensor level, Hamamatsu specifies the R12421 as a compact 13 mm diameter photomultiplier tube with a 10 mm effective area, a bialkali photocathode, 10 dynode stages, a spectral response from 300 to 650 nm, and a peak response near 420 nm [17]. The same datasheet gives a typical gain of 2.0×10^6 at 1000 V, an anode pulse rise time of about 1.2 ns, and a maximum anode-to-cathode voltage magnitude of 1250 V [17]. This spectral response is technically compatible with stilbene emission in the blue-violet region, so the R12421 is a consistent readout choice for a compact stilbene monitor operated without an intermediate wavelength-shifting stage.

For the present campaign, current working notes indicate direct raw-anode readout with negative polarity and negative high-voltage operation of the assembled detector. The working pulses are described as several hundred nanoseconds wide and typically in the few-hundred-mV range into $50\,\Omega$, with amplitudes potentially approaching about 1 V at operating bias. Because the factory sheet for the exact SCIONIX assembly is not yet available locally, these signal-level statements should be interpreted as setup-specific working values rather than universal device specifications.

The most defensible experimental role of this detector in the present setup is therefore that of a relative neutron monitor and beam-reference channel. The crystal is intentionally compact, and the Scionix detector note explicitly remarks that efficient fast-neutron detection requires a significant absorption volume [24]. Accordingly, the probe is well suited to tracking relative beam conditions and providing a compact PSD-capable reference detector in parallel with the main NCAL measurement, but it should not be treated in this report as an absolutely calibrated neutron-efficiency instrument unless a separate dedicated calibration is performed.

Table 4.5: Selected characteristics of the facility-provided stilbene monitor.

Parameter	Value	Comment
Assembly identifier	SCIONIX `V5B15/0.5-E1-STILB-X-NEG`	Control-detector identification from campaign notes.
Scintillator material	Crystalline stilbene	PSD-capable organic fast-neutron scintillator [19, 24].
Nominal crystal size	$5 \times 5 \times 15$ mm	Size recorded for the assembled probe in campaign notes.
Stilbene density	$1.15\,\mathrm{g\,cm^{-3}}$	Inrad material value [19].
Maximum emission wavelength	$380\,\mathrm{nm}$	Inrad material value [19].
Prompt decay time	About $4.3\,\mathrm{ns}$	Inrad material value relevant to fast timing [19].
Photosensor	Hamamatsu `R12421` PMT	Small PMT documented by Hamamatsu [17].
PMT effective area	10 mm	Hamamatsu datasheet value [17].
PMT spectral response	300 to 650 nm, peak near $420\,\mathrm{nm}$	Compatible with stilbene blue-violet emission [17].
Typical PMT gain	2.0×10^6 at $1000\,\mathrm{V}$	Hamamatsu datasheet value [17].
Anode pulse rise time	About $1.2\,\mathrm{ns}$	Hamamatsu datasheet value [17].
PMT voltage limit	$\lvert V_{\mathrm{AK}}\rvert \leq 1250\,\mathrm{V}$	Tube-level maximum rating [17].
Campaign readout branch	Raw anode output	Setup-specific operating choice.
Observed / expected signal form	Negative pulses, several hundred ns wide, typically few hundred mV into $50\,\Omega$, possibly approaching $1\,\mathrm{V}$	Setup-specific working values, not factory-guaranteed specifications.
Intended role in this campaign	Relative neutron monitor and beam-flux reference	Compact parallel reference channel for the NCAL run.

Chapter 5

Data-Acquisition Systems

This chapter describes the front-end and support acquisition systems. The offline reconstruction and physics-oriented interpretation are treated separately in Chapter 6.

5.1 CAEN ADC DT5725S

The first data-acquisition stage of the present NCAL commissioning campaign was based on a CAEN DT5725S desktop digitizer. In the public CAEN product documentation, the DT5725 and DT5725S are grouped into the same instrument class: an 8-channel, 14-bit, $250\,\mathrm{MS/s}$ digitizer of the x725 family [10]. After the Linux USB driver had been installed on the laboratory computer, a direct CAEN library probe identified the board used in this work as model DT5725S, with 8 channels and 14-bit conversion. The same probe also showed that the board was already configured with a DPP-PSD application firmware. For that reason, the first functional test was performed with *CoMPASS*, rather than by immediately switching to a waveform-only workflow.

An important feature of the x725 platform is the use of a programmable FPGA rather than a single fixed acquisition logic. The same hardware can therefore be operated either as a waveform recorder or as a digital pulse processor, depending on the firmware image loaded onto the board [10, 11]. This flexibility is highly relevant for detector-development work. If the main objective is to preserve the full sampled pulse for later offline analysis, then waveform recording is the natural mode. If, instead, the aim is online extraction of energy, timing, or pulse-shape observables with a smaller event size, then one of the DPP firmware families is generally more appropriate.

For practical operation, the S suffix must be treated carefully. CAEN presents the DT5725 and DT5725S together at product level, but the downloadable firmware packages are distributed separately for x725 and x725S devices. In that sense, the DT5725S should be understood as the S-series hardware revision of the same 8-channel $250\,\mathrm{MS/s}$ digitizer platform: the overall DAQ role is unchanged, but the correct x725S firmware image must be used for upgrades or mode changes. This distinction is operationally important even when the high-level acquisition concept remains the same.

5.1.1 Firmware families and operating modes

The firmware choice determines what information is extracted online and how much data reduction is performed before storage. The main modes relevant to the DT5725S are summarized in Table 5.1.

In the present commissioning, the installed firmware belongs to the DPP-PSD family, which is well aligned with fast first tests using scintillation detectors and PMT-like signals.

Table 5.1: Firmware modes relevant to the DT5725S/x725S platform.

Firmware	Main operating principle	Typical use case
Waveform Recording (D-WAVE)	Oscilloscope-like waveform acquisition based on circular buffers, time-windowed recording, and dead-timeless multi-buffer management [11]	Detector R&D, pulse-shape studies, charge integration, baseline studies, and flexible offline analysis of raw traces.
DPP-PHA	Digital replacement of a shaping amplifier plus peak-sensing ADC for pulse-height analysis [6]	Gamma spectroscopy and precision energy measurements with HPGe, silicon detectors, or scintillators coupled to charge-sensitive front-end electronics.
DPP-PSD	Digital replacement of a dual-gate QDC, discriminator, and gate generator, with charge integration and pulse-shape discrimination [7]	Scintillator readout, especially when neutron–gamma discrimination or fast timing is required. This is the firmware presently installed on the laboratory board.
DPP-DAW	Self-triggered acquisition with dynamic record length adjusted to the pulse duration, optimized for zero suppression and trigger-less operation [5]	Sparse-data systems, variable pulse lengths, and applications where efficient data reduction is important.
DPP-ZLEplus	Compressed waveform transfer with enhanced zero suppression and programmable look-back/look-forward windows around significant pulses [9]	Acquisition scenarios where waveform fragments should be preserved, but the volume of baseline-only data must be reduced.
DPP-SUP	License bundle covering several DPP algorithms on a single supported digitizer [8]	Convenient when the same hardware may need to switch between multiple DPP modes during development or deployment.

From a DAQ-design perspective, these firmware families correspond to different compromises between information content and throughput. Waveform recording maximizes retained information, because the full sampled trace is available for later analysis. The DPP modes instead move part of the signal processing into the FPGA and store higher-level observables such as integrated charge, pulse height, timing, or discrimination parameters. This reduces the event size and simplifies online monitoring, but it also couples the recorded data more strongly to the assumptions built into the chosen algorithm. For the first commissioning step, keeping the already installed DPP-PSD firmware was therefore the most conservative choice: it avoided an unnecessary firmware change while allowing immediate validation of the readout chain.

5.1.2 Driver requirement on the acquisition computer

For Linux-based operation, USB visibility at the operating-system level is not sufficient by itself. CAEN applications for Digitizer 1.0 boards rely on the CAEN software stack, including the communication and digitizer libraries used by programs such as WaveDump and CoMPASS [3, 14]. In the present commissioning, the DT5725S was initially visible through the generic USB subsystem, but the CAEN libraries were still unable to open the device. Only after installation of the CAENUSBdrvB Linux driver did the expected device interface `/dev/usb/v1718_0` appear; after that step, the same probe opened the board successfully and read back the model and firmware information.

This observation explains why the driver is mandatory on the acquisition computer. It does not merely provide user permissions for an already usable USB endpoint; rather, it supplies the device interface expected by the CAEN communication layer. In practical terms, the successful commissioning sequence on the laboratory PC was therefore: USB driver installation, verification of the `v1718` device node, successful low-level library access, and only then use of the graphical DAQ software.

5.1.3 Software environment

The DT5725S is embedded in a broader vendor software ecosystem, and different programs are appropriate for different firmware modes. The most relevant tools for the present work are summarized in Table 5.2.

Table 5.2: Main software components relevant to the DT5725S.

Software	Primary role	Relevance to the present setup
CoMPASS	Multiparametric DAQ for CAEN DPP firmware, with simultaneous waveform, energy, timing, PSD, and list-mode handling [4]	Natural front-end for the current DPP-PSD board configuration; used for the first successful functional test after driver installation.
WaveDump	Basic console readout program for boards running waveform-recording firmware; supports configuration files, rate readout, waveform display, and data export [14]	Useful as a lightweight reference application and as an example of direct use of the CAEN libraries. It does not support DPP firmware.
WaveDump2	Modern open-source GUI for waveform-recording operation, with oscilloscope-like plots, multi-board handling, configuration save/load, and ASCII or binary output [13]	Most relevant if the DT5725S is later reflashed from DPP-PSD to x725S waveform-recording firmware for raw-trace studies.
CAEN Toolbox	Vendor utility for firmware upgrades, license management, PLL handling, and low-level board debugging [12]	Relevant when changing firmware family, updating the board, or managing licenses during later stages of the DAQ campaign.

Practical launch commands. For Linux-based operation, the most useful user-level launch commands are straightforward. The DPP-oriented graphical front-end can be started as `CoMPASS`; if no shell launcher has been defined, the equivalent vendor-bundle invocation is `./CoMPASS` from the installation directory. The legacy console waveform reader is typically started as `WaveDump <configuration-file>`. The modern waveform GUI is started as `WaveDump2`, and the firmware and license-management utility is started as `caen-toolbox`. In practical use, the exact capitalization of these commands matters.

Two further remarks are important. First, the waveform-oriented tools are distinct from the DPP-oriented tools. The legacy WaveDump program and the more recent open-source WaveDump2 are intended for waveform-recording operation, whereas CoMPASS is designed around CAEN DPP algorithms [4, 13, 14]. Second, CAEN also refers to CAENScope as a waveform-control environment in the waveform-recording documentation [11]; in the present work, however, the practically relevant distinction is between the current DPP-PSD plus CoMPASS path and a possible future x725S waveform-recording plus WaveDump2 path.

5.1.4 Reproducible laboratory procedure

To avoid repeating the same transport-layer debugging during later shifts, the first functional test should follow a fixed and documented sequence. On the present acquisition PC, the CAENUSBdrvB package was unpacked, built, and installed directly with `make` followed by `sudo make install`; this was the path actually used because a DKMS-based installation was not available on the host at that time. After installation, the decisive check was the appearance of the expected CAEN device node `/dev/usb/v1718_0`.

For subsequent laboratory work, the minimum reproducible procedure is the following.

1. Verify that the digitizer is powered, connected by USB, and visible to the operating system as a CAEN device.

2. Confirm that the CAENUSBdrvB driver is installed for the running kernel and that the `v1718` device node is present.

3. Run a low-level CAEN access test, or an equivalent vendor utility, and verify that the board opens successfully and reports the expected model identifier `DT5725S`, channel count, and firmware information.

4. Start *CoMPASS*, add the board as a direct USB connection with link number 0, and keep the board in its currently installed firmware family during the first validation step.

5. Enable one detector channel first, apply a conservative threshold and the correct signal polarity, and verify that both the trigger rate and the displayed waveform respond when a detector signal, source, or pulser is present.

6. Archive the accepted setup by saving the *CoMPASS* configuration together with the date, serial number, firmware revision, driver package version, active channel list, and the location of the first validated data file.

This sequence is intentionally modest, but it is sufficient to distinguish a solved infrastructure problem from a merely temporary success. A run should be treated as a valid commissioning reference only after the driver interface, low-level CAEN communication, graphical DAQ connection, and single-channel pulse response have all been demonstrated in the same session.

In summary, the present commissioning strategy was scientifically and technically motivated by the actual state of the hardware. The board already carried a DPP-PSD firmware, and the first bottleneck on the lab PC was not the firmware itself but the missing Linux USB driver. Once that driver was installed and the CAEN library stack could open the device, CoMPASS became the shortest path to a first validated acquisition. A later transition to full raw-waveform studies remains fully possible, but it would require loading the x725S waveform-recording firmware and switching to a waveform-oriented software stack.

5.2 DOGMA edge-stream readout

The time-of-flight-oriented beam dataset analyzed in the present work is acquired with the DOGMA chain rather than with the CAEN DPP list-mode path. In this chain the archived event model is edge based. The raw files store leading and trailing edge times, and the later pulse quantities such as rise time and time-over-threshold are reconstructed offline. The present repository therefore begins from DOGMA-produced raw files with names such as `NCAL_..._.dld.dat`. The upstream conversion to that format is outside the scope of the current document.

A defining feature of the DOGMA dataset is the dedicated timing reference on channel 0. The machine-related RF signal is discriminated, prescaled, and delivered both to the trigger path and to `ch0`, so the recorded edge stream already contains the reference needed for precise relative-timing studies. This DAQ architecture is one of the main reasons why the DOGMA data support the later folded-RF and time-of-flight analyses more naturally than a trigger-time-only event record would.

5.3 Oscilloscope support diagnostics

Although the oscilloscope is not the primary archival DAQ for the present physics dataset, it remains an essential commissioning instrument. Direct waveform inspection is the fastest way to verify signal polarity, baseline stability, gross saturation, reflections, light leaks, and cable integrity before a long run is accepted. In practice, oscilloscope observations provide the bridge between detector hardware and the digitized data stream: they show whether the analog pulse is physically plausible before it is reduced either by a DPP firmware or by the DOGMA edge extractor.

For that reason, oscilloscope measurements are treated here as laboratory validation tools rather than as a separate production-data stream. Whenever a later anomaly appears in rate, timing, or time-over-threshold observables, a direct waveform check remains the first low-level cross-check.

5.4 DiRICH comparison context

Earlier NCAL development campaigns also used a DiRICH-based readout chain, and that historical experience remains scientifically relevant because it provides a comparison point for timing-reference quality, signal handling, data integrity, and downstream analysis burden. In the present working document, however, the active curated dataset and all currently synchronized analysis products are DOGMA based. The detailed offline methodology is therefore presented in the next chapter for the DOGMA edge-stream campaign, while DiRICH is retained here as a DAQ-system context and as a benchmark for future comparative studies.

Chapter 6

Analysis

This chapter groups the current analysis paths by acquisition chain. The synchronized beamtime workflow is presently DOGMA based, so the DOGMA section is the scientific core of the chapter. The CAEN, oscilloscope, and DiRICH sections are included for methodological completeness and for future comparative development.

6.1 DOGMA analysis

This section describes the offline methodology applied after the DOGMA edge-stream data have been recorded. It is separated deliberately from the DAQ-systems chapter because the scientific logic of pulse reconstruction, timing-reference control, RF folding, and gamma–neutron separation is not identical to the hardware logic of the front-end electronics.

The active production code is maintained in the repository analysis directory, while exploratory or superseded utilities are moved to the adjacent archive area. Code is edited locally in the mirrored workspace, but authoritative processing is executed remotely on `padiwa` against the canonical `/data6` filesystem. This split preserves version control locally. It also keeps the heavy computation and the large result trees on the storage system for which the workflow was designed.

The present repository begins from DOGMA raw edge files with names such as `NCAL_..._.dld.dat`. Upstream conversion to this edge-stream format is external to the current analysis repository; the workflow described here begins once those files already exist.

For reproducibility, the main storage layout is fixed.

- Raw DOGMA inputs are stored under `/data6/Data/` as `*.dld.dat` files.
- Cleaned pulse tables are stored under `/data6/Data/Refined_Data/`.
- Per-run analysis products are stored under the results root `/data6/Dogma_analysis_by_Dachi/Results/`, with one subdirectory per run key.
- Batch-comparison plots are collected in the `Ch02_Position_Comparison/` subtree of that same results root.
- Archived result sets are kept under `Results/Archive/`, with archived cleaned-data directories under `/data6/Data/Refined_Data/Archive/`.

The analysis uses a compact *run key* derived from the raw filename, for example `NCAL_20us_Pos_3.4m_0000`. This run key is then used consistently for refined directories, result directories, summary tables, ROOT outputs, and comparison plots. The convention is operationally important because it allows regenerated products to be compared run by run rather than by full raw filename.

6.1.1 Active step map

Table 6.1 summarizes the currently available production and diagnostic steps in the active DOGMA chain.

Table 6.1: Current DOGMA cleaned-data workflow and analysis diagnostics.

Stage	Main code / primary outputs	Scientific role
01	`step01_dogma_raw_data_refiner.cpp`, cleaned pulse TSV table	Rise–fall pairing, trigger-window grouping, basic acceptance cuts, and creation of the canonical pulse-level intermediate product.
01R	`step01_write_cleaned_root_tree.py`, `CleanedHits` ROOT export	Final reusable pulse-level ROOT representation with timing and RF metadata for downstream studies.
02	`step02_*`, all-channel rate summaries and plots	Rate monitoring, run-level sanity checks, and absolute-rate comparisons across channels and runs.
03	`step03_*`, trigger-ref and `ch0`-ref time–ToT products plus whole-run ToT distributions	Time structure, pulse-width distributions, and validation of the timing-reference choice.
04	`step04_*`, RF-period scans, folded profiles, folded phase–ToT products, and RF segment tables	Deduction of the RF period, phase folding, and robust handling of within-run RF resets.
05	`step05_estimate_gamma_neutron_background.py`, sideband background summaries	First-pass local background estimates in phase–ToT space.
06	`step06_quantify_gamma_neutron_components.py`, ROI overlays, projections, and component tables	Quantification of prompt-gamma and constrained fast-neutron candidate regions with explicit overlap flags.
07	`step07_phase_rate_slice_diagnostics.py`, one-period phase-rate slices and excess intervals	One-dimensional delayed-excess diagnostics before finalizing two-dimensional selections.
08	`step08_fit_gamma_tot_peak.py`, gamma-ToT peak-fit tables and figures	Run-by-run centering of the compact prompt-gamma ToT feature in a narrow fit interval.
09	`step09_gamma_window_sensitivity.py`, width-sensitivity summaries	Systematic study of prompt-gamma reference stability under different centered ToT-gate widths.
10	`step10_gamma_reference_cfd.py`, reference-comparison diagnostics	Comparison of local-fit, raw-maximum, and CFD-style prompt-gamma reference definitions.
Batch	`run_all_cleaned_analysis_chain.py`, synchronized per-run result trees	Reproducible remote orchestration of the cleaned-data production chain.

The numerically heavy transformations remain in compiled C++ code, whereas Python is used mainly for orchestration, file management, bookkeeping, and rendering. This separation is deliberate: the workflow needs compiled performance for long runs, but it also needs enough scripting flexibility to regenerate summaries, diagnostics, and comparison plots without rewriting the analyzers themselves.

6.1.2 Pulse reconstruction and timing observables

Pulse cleaning as the canonical intermediate

The first production step is implemented in `step01_dogma_raw_data_refiner.cpp` and driven by `step01_run_dogma_raw_data_refiner.py`. Its purpose is to convert raw edge information into a cleaned pulse table. Conceptually, the raw DOGMA file contains edge-level information, whereas the later physics studies require pulse-level quantities. The cleaner therefore groups edges into trigger windows, matches rising and falling edges to form pulses with well-defined rise time, fall time, and time-over-threshold (ToT), rejects isolated or inconsistent edges, applies configurable acceptance limits, and writes run-level summary files so that anomalies remain auditable.

This step is also the natural place to guard against global-time discontinuities. The refined pulse table is therefore treated as the canonical reusable intermediate product: once it exists and has passed sanity checks, most later analysis stages can be regenerated without rereading the original raw edge file. The final ROOT writer, `step01_write_cleaned_root_tree.py`, consumes the same cleaned pulse table together with the RF-segmentation metadata and writes the split `CleanedHits` export used for later reuse.

Rate, ToT, and timing-reference products

After pulse cleaning, `step02_*` writes cleaned all-channel rate products, while `step03_*` writes trigger-referenced and `ch0`-referenced time–ToT products together with per-channel ToT distributions. These outputs are scientifically useful for two reasons. First, they verify that the cleaned pulse table is populated by accepted pulse candidates rather than by unmatched edges. Second, they expose the prompt-like pulse-width feature that later becomes the anchor for the gamma reference.

Two timing reference frames are used throughout the cleaned-data chain.

1. **Trigger-referenced time**, where pulse times are measured relative to the hardware trigger.
2. **Channel-0-referenced time**, where pulse times are measured relative to the accepted reference pulse on channel 0 within the same trigger window.

The `ch0` reference is physically important because channel 0 carries the discriminated timing signal derived from the machine RF chain. In the current implementation, a `ch0` candidate is accepted only if its rise time and ToT fall inside a predefined validity window. If several valid candidates are present, the first valid one is selected, and companion statistics files are written so that this choice can be audited later. For a non-reference channel, the key timing quantity is therefore

$$t_{\mathrm{ch0\,ref}} = t_{\mathrm{rise}}^{(\mathrm{ch})} - t_{\mathrm{rise}}^{(\mathrm{ch0})}. \tag{6.1}$$

Figure 6.1 shows a representative whole-run channel-2 ToT spectrum from the cleaned pulse table. The narrow peak near 13.5 ns is the compact prompt-like feature later isolated by the step-08 fit, while the broad lower-ToT and higher-ToT structures demonstrate why a narrow, dynamically centered prompt-gamma gate is preferable to a fixed hard-coded ToT interval.

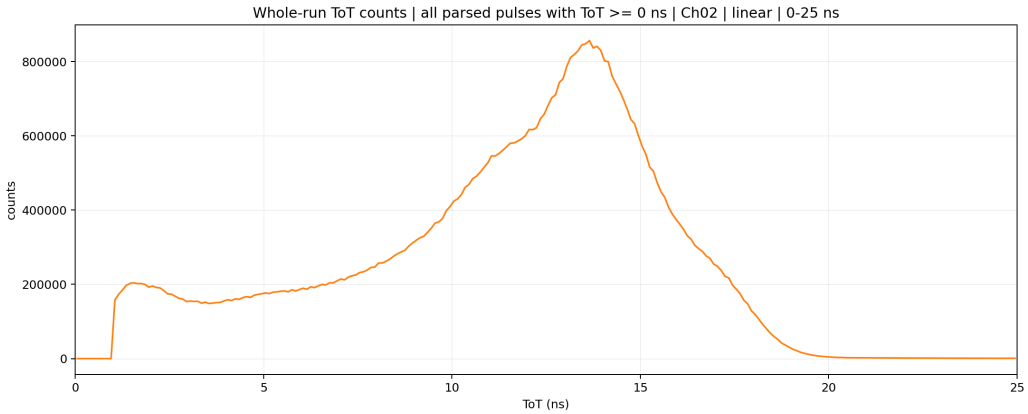


Figure 6.1: Representative whole-run cleaned channel-2 ToT distribution for the 5.0 m position run. The spectrum is built from accepted pulses after edge pairing and cleaning. The sharp feature near 13.5 ns is the compact prompt-like ToT component later used for run-by-run gamma-gate centering, whereas the broad surrounding structure motivates a two-stage prompt-reference treatment rather than a fixed global cut.

RF period deduction and folded observables

Measured millisecond trigger envelope. The fast RF period must be distinguished from a slower modulation of the recorded trigger population. A cyclotron-control-room oscilloscope photograph supplied during the timing investigation shows an approximately 1 ms per division timebase, a pattern repeating approximately every 6.6 ms to 6.7 ms, and a short interval of about 1.2 ms to 1.4 ms associated with bursts on the second trace (Figure 6.2). This is consistent with the operational recollection of approximately 150 Hz modulation and approximately 20% active duty. The photograph motivates that hypothesis, but does not by itself identify the scope-channel wiring, active polarity, or an exact beam-current duty cycle.

The hypothesis was tested independently of detector-pulse and timing-reference cuts using every original `TDC_001049` trigger-window header in a fresh decode of the canonical 5.0 m `_0000` recording. All 10,516,211 timestamps were retained, including windows without a passing cleaned `ch0` reference. The original `GlobalTriggerTime` values were mapped to their native 5 ns integer-counter lattice before differences were formed. Their absolute DAQ-time range is $902.178\,410\,820$ s to $2952.996\,736\,650$ s, spanning $2050.818\,325\,830$ s; these are DAQ counter times, not UTC. Figure 6.3 shows the full distribution and unfiltered early, middle, and late examples.

For envelope measurement only, consecutive timestamps separated by at most 0.2 ms were grouped and groups with at least eight triggers were labelled dense; sparse groups were not

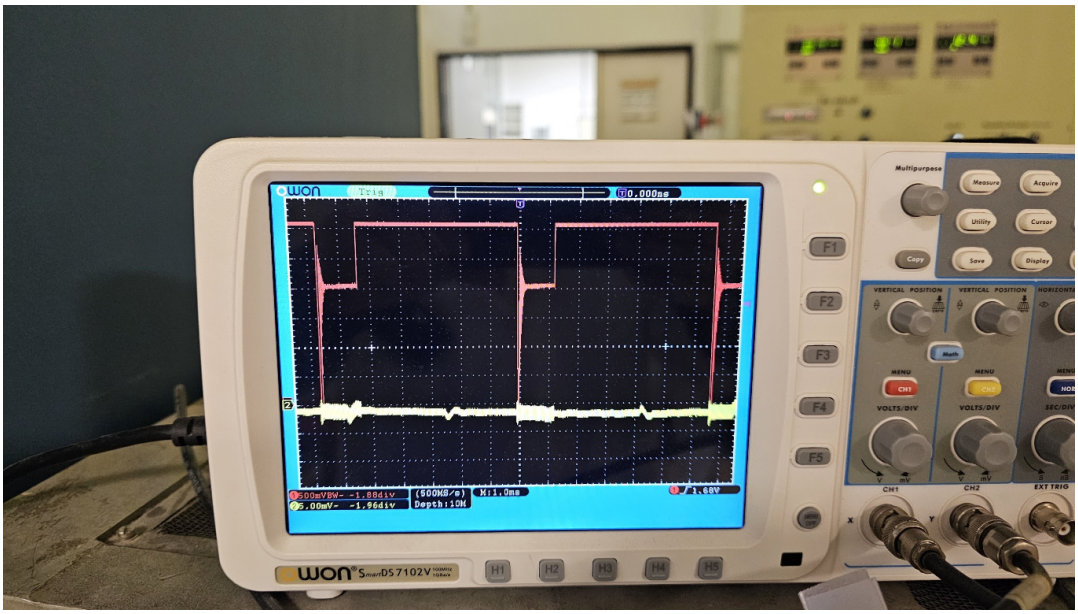


Figure 6.2: Cyclotron-control-room oscilloscope photograph supplied as timing context. The short interval occupies approximately one fifth of the repeating millisecond-scale pattern, and the yellow bursts appear associated with the low portion of the pink trace. Individual RF cycles are not resolved at this timebase. Trace identities and the hardware gating mechanism are not inferred from the photograph alone.

removed from the absolute distribution or accounting. This identifies 307,491 dense bursts containing 10,458,943 windows, or approximately 99.455% of the recording, alongside 57,268 sparse-group windows. Dense bursts typically contain 33–35 triggers, with median within-group spacing $39.780\,\mu s$. Their median start-to-start spacing is $6.665\,070\,ms$. Regression in blocks of at most 256 burst starts, without assigning cycle counts across nonregular gaps, gives an aggregate repetition period of approximately $6.665\,156\,ms$, or $150.034\,Hz$. No $150\,Hz$ template or 20% duty selection was imposed.

The median observed first-to-last trigger span is $1.312\,830\,ms$. Since those first and last triggers need not coincide with the envelope boundaries, a recorded-trigger occupancy proxy is defined by

$$d_{\text{trig}} = \frac{t_{\text{last}} - t_{\text{first}} + \widetilde{\Delta t}_{\text{within}}}{t_{\text{first,next}} - t_{\text{first}}}, \tag{6.2}$$

where $\widetilde{\Delta t}_{\text{within}}$ is the median within-group trigger spacing. This gives a median proxy of 20.29%, with a central 68% range of 19.78–20.81%; the median corrected span is $1.352\,610\,ms$. The one-spacing endpoint convention is about 0.6 percentage point on this cycle, so these values are not an exact independently measured RF-on gate or beam-current duty. The median separation from the last trigger of a dense burst to the first of the next is $5.352\,940\,ms$; sparse triggers can occur inside that separation.

The feature persists throughout ten equal-duration recording segments, whose fitted repetition

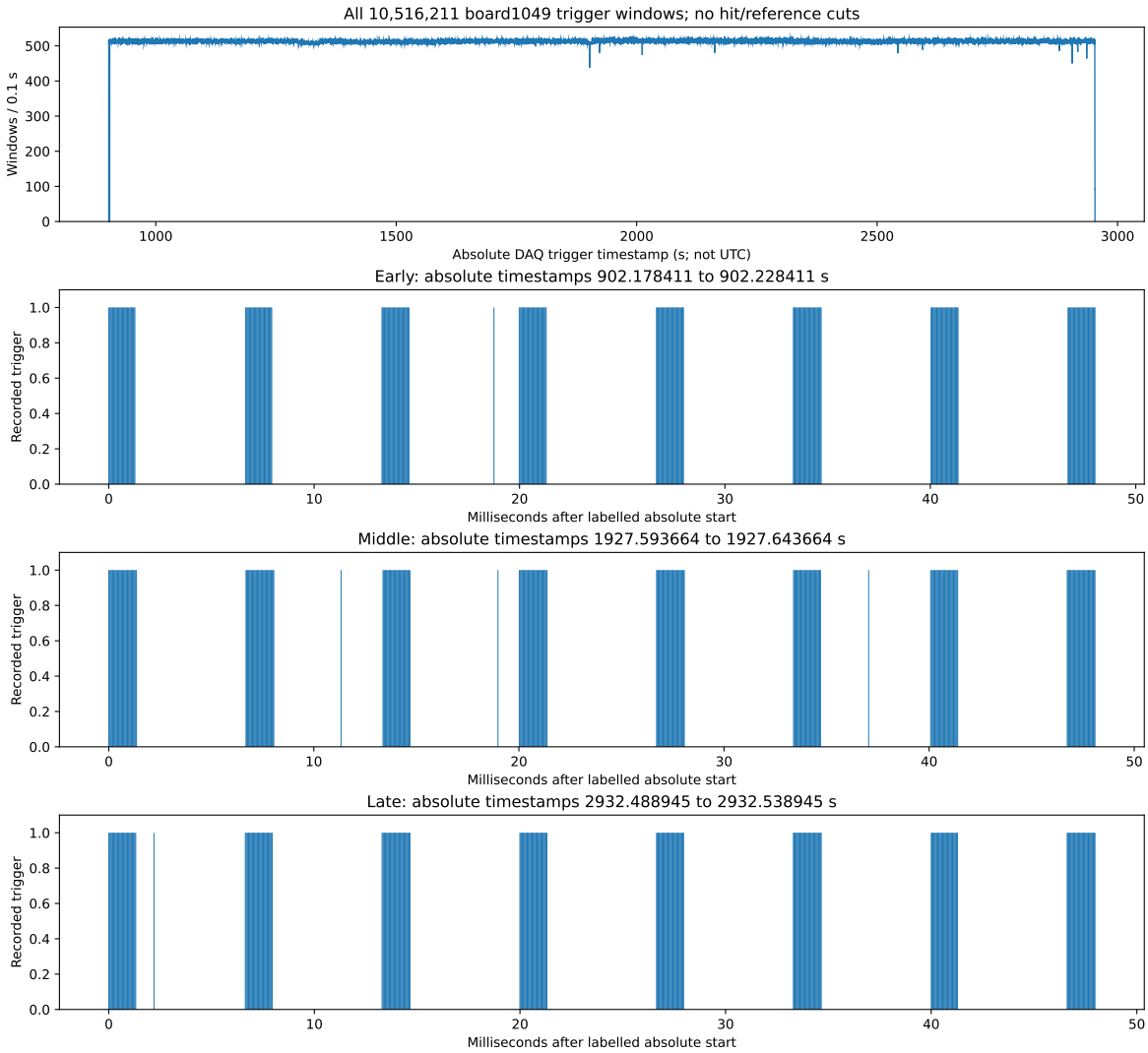


Figure 6.3: Absolute DAQ trigger-window timestamp distribution for all 10,516,211 board1049 windows, without edge, ToT, `ch0`-reference, or clock-support cuts. The upper panel uses 0.1 s bins over the recording. The lower panels retain every trigger in three 50 ms intervals identified by their absolute starting timestamps. Dense bursts approximately 1.3 ms wide repeat on a 6.67 ms timescale; sparse inter-burst triggers are also visible.

frequencies range from 149.817 Hz to 150.234 Hz and whose median duty proxies range from 20.267% to 20.316%. Grouping thresholds from 0.1 ms to 1.0 ms reproduce the same dense-burst count and give frequencies approximately 150.034 Hz to 150.068 Hz; density classifications were also varied. These spreads describe sensitivity and observed variation, not statistical confidence intervals. All absolute histogram bins were independently checked with integer counters, and 10,518 sampled original timestamp strings were independently checked by decimal arithmetic.

The approximately 150 Hz, approximately 20% recorded-trigger envelope is therefore an established feature of this recording, consistent with the scope image. The physical interpretation as modulation of the RF-derived timing chain is supported, but timestamps alone do not distinguish RF-amplitude suppression, discriminator threshold effects, trigger gating, or actual carrier interruption. Sparse triggers are explicitly retained, and a separate $1.280\,333\,970$ s inter-trigger gap remains outside the ordinary modulation explanation. Other runs require their own checks.

Consequently, ordinary inter-burst gaps must not automatically be labelled data loss, corrupted time, machine restarts, or RF phase resets. Nor should otherwise valid detector hits be discarded merely because a symmetric local-period estimator lacks neighbours at a natural burst boundary. The earlier gap-aware ch0 diagnostic imposed such a support selection and is a selected-subset diagnostic, not a matched-population alternative to the original ch2 estimator. Period-estimator comparisons must use the same cleaned, referenced detector hits and explicitly state any constant-period or cross-burst phase-continuity assumptions. This timestamp investigation changes no existing pulse-selection or folded-histogram product.

The RF-folding step, implemented in `step04_*`, deduces a best RF period P and a phase origin ϕ_0 from a score channel, which is currently channel 2 by default. The folded phase is written using the `ch0`-referenced rise time,

$$\phi = \mathrm{mod}(t_{\mathrm{ch0\,ref}} - \phi_0,\ P)\,. \tag{6.3}$$

This distinction is physically essential: trigger-referenced times and channel-0-referenced times are both useful, but the folded RF products are intended to characterize detector timing relative to the RF reference, so the meaningful phase coordinate is the channel-0-referenced one.

The current robust version of the workflow also supports within-run RF segmentation. If the cyclotron is stopped and restarted during one run, one global phase origin is no longer sufficient. The patched step-04 analyzer therefore searches time-ordered score pulses for persistent phase jumps, writes a phase-segment table, and passes that segmentation to the final ROOT writer through the `rf_segment_id` branch. In this way, a restart no longer forces the entire run to share one RF phase origin.

6.1.3 Robustness controls and diagnosed failure modes

Several conceptually different problems were encountered during commissioning, and they materially shaped the current workflow.

Edge pairing and `ch0` ambiguity. The first class of questions concerned whether the cleaner was pairing edges in a physically sensible way and how robust the `ch0` reference selection remained when multiple candidates appeared in the same trigger window. The practical response was to treat the pulse cleaner as the authoritative rise–fall pairing stage and to extend the runners so that odd-`ch0` statistics and sample files are written explicitly. Ambiguities can therefore be inspected directly instead of being inferred indirectly from final plots.

RF phase–ToT combs. The folded RF phase-versus-ToT plots showed persistent negative-slope diagonal structures. The decisive diagnostic, using `phi_star_artifact_diagnostic.py` together with direct comparisons between pre-fold and post-fold outputs, showed that these structures are not introduced by the rendering or folding stage itself. The software can sharpen them, but it is not their primary origin. The working interpretation is therefore that the reference or signal chain contains preferred timing families that become visible once the data are folded.

Reference-chain interpretation. Once the software-artifact hypothesis became unlikely, attention shifted to the hardware timing chain. In particular, the RF sine is converted to logic through a discriminator path, so threshold-crossing families in that chain remain a plausible source of structured timing bunching. In this interpretation, the analysis code is correctly exposing a property of the timing-reference chain rather than inventing one.

Reproducibility of long reruns. For the large runs, the workflow cost is not dominated solely by the initial cleaner. RF folding and final ROOT export can each consume substantial wall time. This motivated two policies that remain central to the present chain: the cleaned pulse table is treated as the stable handoff between raw edge decoding and higher-level timing analysis, and regenerated remote products are archived run-key by run-key rather than overwritten without trace.

6.1.4 Gamma–neutron component studies

The folded RF phase-versus-ToT plots are useful for developing gamma–neutron separation strategies, but they should not be interpreted as direct particle-identification maps without a background model. The radiation field seen by NCAL is mixed. It can contain prompt gammas from source-station materials, secondary capture gammas, room and shielding backgrounds, scattered neutrons, continuum neutrons from the beryllium target, random coincidences, electronic pickup, and residual timing-chain artifacts. For this reason, a compact region in phase–ToT space should be called

gamma-like or neutron-like only after the non-beam and off-region contributions have been characterized.

The working strategy is hierarchical. First, beam-off, blocked-aperture, or sideband regions should be used where available to estimate components that are not correlated with prompt beam timing. Second, cleanly separated regions of the position scan can be used to define empirical gamma-like and neutron-like templates. Third, regions where the two populations overlap should be treated statistically rather than by hard event labels. In an overlapped phase–ToT region, the observed density is represented schematically as

$$N(\phi, \mathrm{ToT}) = A_\gamma G(\phi, \mathrm{ToT}) + A_n N_n(\phi, \mathrm{ToT}) + B(\phi, \mathrm{ToT}), \tag{6.4}$$

where G is the prompt-gamma-like template, N_n is the neutron-like template, and B is the residual background model. Under such conditions, individual hits should not be assigned uniquely to gamma or neutron populations using phase and ToT alone; they should instead receive probabilistic weights or appear only in fitted yield estimates with explicit overlap uncertainty.

An analytical time-of-flight reference is also needed to show why the position scan is expected to help. For a detector-face distance d, the prompt-gamma time is

$$t_\gamma(d) = \frac{d}{c}, \tag{6.5}$$

while a neutron with kinetic energy E_n has

$$\beta_n(E_n) = \sqrt{1 - \left(1 + \frac{E_n}{m_n c^2}\right)^{-2}}, \qquad t_n(d, E_n) = \frac{d}{\beta_n c}. \tag{6.6}$$

The reference separation is therefore

$$\Delta t(d, E_n) = t_n(d, E_n) - t_\gamma(d). \tag{6.7}$$

For the present campaign, 30 MeV is used as the main reference energy because the measured forward proton-on-beryllium spectra at NPI show prominent high-energy structure near 30 MeV to 32 MeV and extend to about 33 MeV for the thick-target configuration [2, 20]. This should be understood as a representative fast-neutron reference curve, not as a monoenergetic source model.

Figure 6.4 shows the resulting prompt-gamma and fast-neutron time-of-flight curves for the position-scan distances. The 30 MeV reference gives separations of about 28.5 ns, 34.6 ns, 40.7 ns, 50.9 ns, 59.0 ns, and 67.2 ns at 2.8 m, 3.4 m, 4.0 m, 5.0 m, 5.8 m and 6.6 m, respectively. Since slower continuum neutrons arrive later, this reference marks the early fast-neutron scale rather than the full neutron timing envelope.

The qualitative motivation for this treatment is visible directly in the folded RF phase-versus-ToT

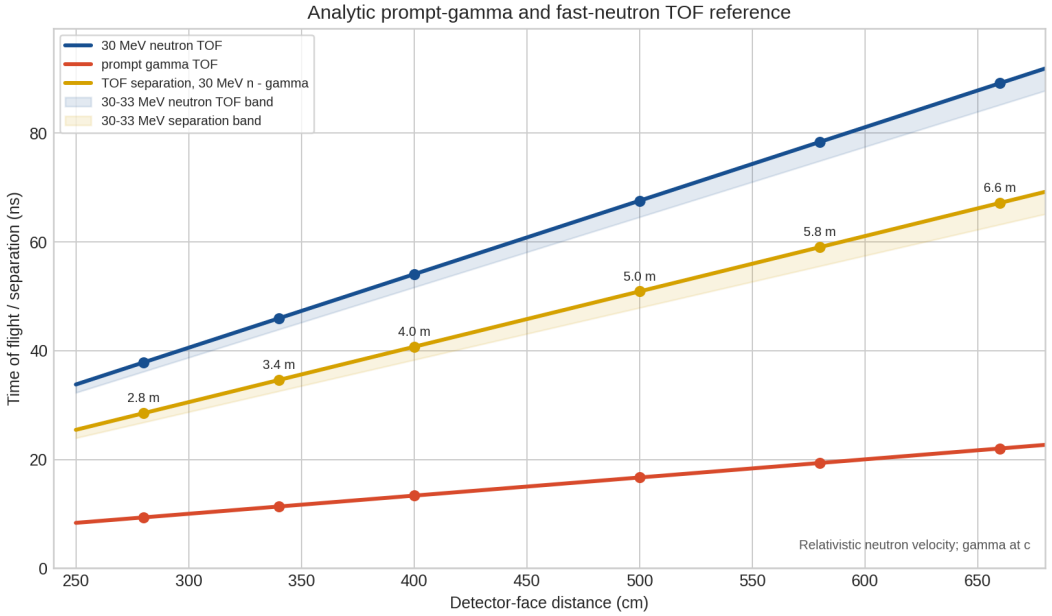


Figure 6.4: Analytical prompt-gamma and fast-neutron time-of-flight reference for the NCAL position scan. The central neutron curve uses 30 MeV; the shaded band indicates the small change between 30 MeV and 33 MeV. The 30 MeV choice is a representative fast-neutron reference within the high-energy part of the 35 MeV proton-on-beryllium spectrum, not a monoenergetic source assumption.

products. Figure 6.5 collects representative channel-2 maps from the completed position-scan runs, with detector-face distance stated explicitly above each panel. The compact prompt-like island remains narrow in phase and ToT, whereas the broader neutron-dominated structure changes its relative timing as the detector distance is varied. The figure is included to show the observable space in which later component modeling must operate; the different panels should not be read as an absolute yield comparison because the displayed density scales are plot-level renderings.

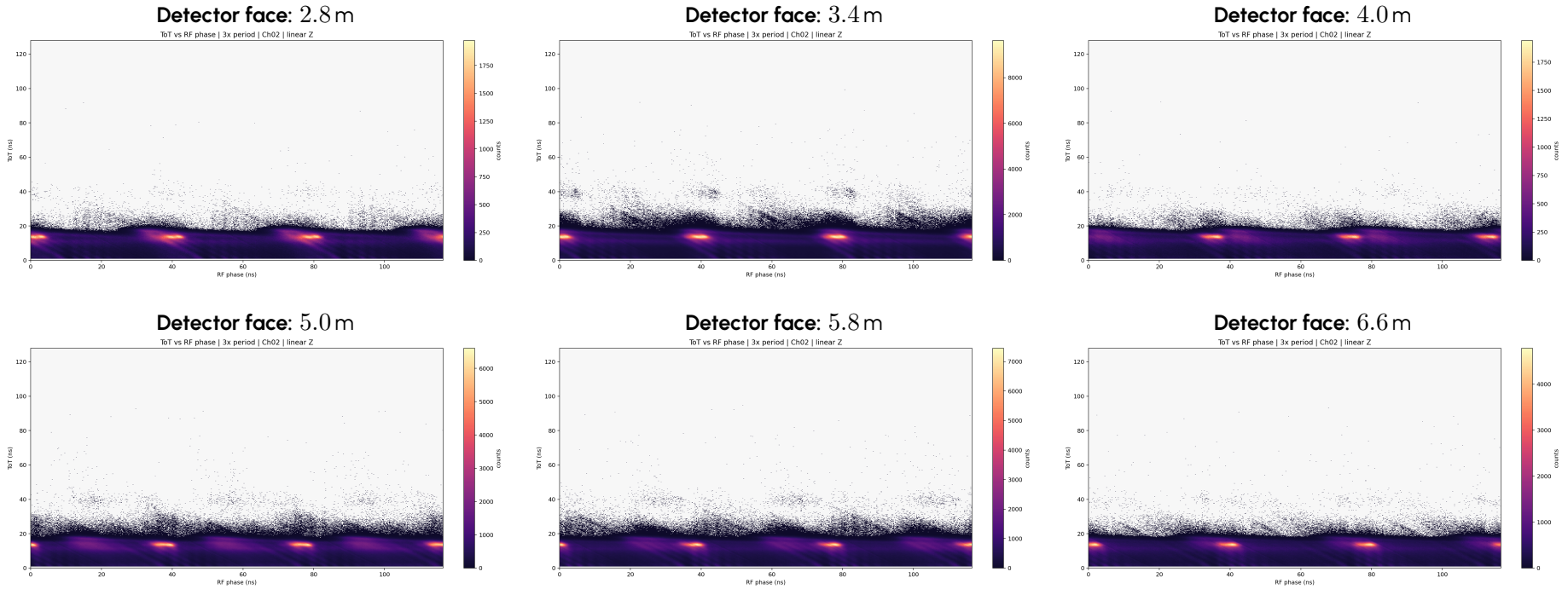


Figure 6.5: Position-scan view of NCAL channel 2 in folded RF phase and time-over-threshold. Each panel shows the same observable space over three RF periods for a different detector-face distance. At some positions the compact prompt-like island is visibly separated from the broader neutron-dominated distribution, whereas in overlap regions the event content must be treated as a statistical mixture rather than by unique hit-by-hit labels. The $4.0\,\mathrm{m}$ panel uses the completed `NCAL_20us_Pos_4m_0001` run.

The first-pass background estimator, `step05_estimate_gamma_neutron_background.py`, uses local sidebands to protect the prompt-gamma and neutron-candidate regions against obvious contamination. The more developed `step06_quantify_gamma_neutron_components.py` then measures prompt-gamma centroids and widths independently for each run, draws the nominal 30 MeV neutron guide, defines protected background sidebands, and reports background-subtracted prompt-gamma and constrained fast-neutron candidate rates. The direct neutron quantity produced by step 06 is intentionally narrow in phase and ToT; it should be interpreted as a fast-neutron TOF/ToT slice rather than as the total neutron yield.

Figure 6.6 shows representative step-06 outputs for the 5.0 m run. The overlay plot displays the compact prompt-gamma ROI, the nominal 30 MeV TOF guide, the observed fast-neutron candidate slice, and the local background sidebands. The corresponding projection plot shows why the step-06 direct neutron number is treated conservatively in overlap cases: the delayed structure is visible and physically plausible, but it is not fully isolated from the prompt-gamma tail.

6.1.5 Prompt-gamma reference construction

The practical challenge in the position scan is not only to identify a gamma-like island, but to anchor every run to a prompt-gamma reference that remains usable when the neutron population is converted into a time-of-flight scale. In overlap configurations the broad neutron-related structure can distort the prompt profile, so a reference extracted directly from a broad phase gate or from a leading-edge CFD construction is not sufficiently stable across all detector positions.

The first dedicated diagnostic is `step07_phase_rate_slice_diagnostics.py`. It collapses selected ToT intervals to one RF period, converts them to rates, and identifies delayed excess intervals relative to a robust off-prompt baseline. This is a selection-design diagnostic rather than a final physics extractor, but it provides an auditable one-dimensional check that the delayed structure is already present before one commits to specific two-dimensional boxes. Figure 6.7 shows the representative 5.0 m case used for this purpose.

The second stage is `step08_fit_gamma_tot_peak.py`, which fits the sharp channel-2 prompt-gamma ToT feature in the fixed interval 12.5 ns to 15.0 ns with a Gaussian plus constant background. Across the completed position runs, the fitted center remains tightly clustered, $\mu_\gamma = 13.580$ ns to 13.665 ns, while the statistical fit uncertainty stays small, $\sigma_\mu \approx 0.0072$ ns to 0.0109 ns. In the present workflow these fit errors are used as a centering precision for the prompt-gamma gate. They are not treated as a complete uncertainty budget for the full ToT distribution, because the broader spectrum is not exhaustively described by a single Gaussian plus constant term.

After the prompt-gamma ToT gate has been centered on μ_γ, the final phase anchor is not taken from a single raw histogram bin. Instead, the cut-band one-period phase profile $r_{\mathrm{cut}}(\phi)$ is formed and a local quadratic model

$$q(\phi) = a\phi^2 + b\phi + c \tag{6.8}$$

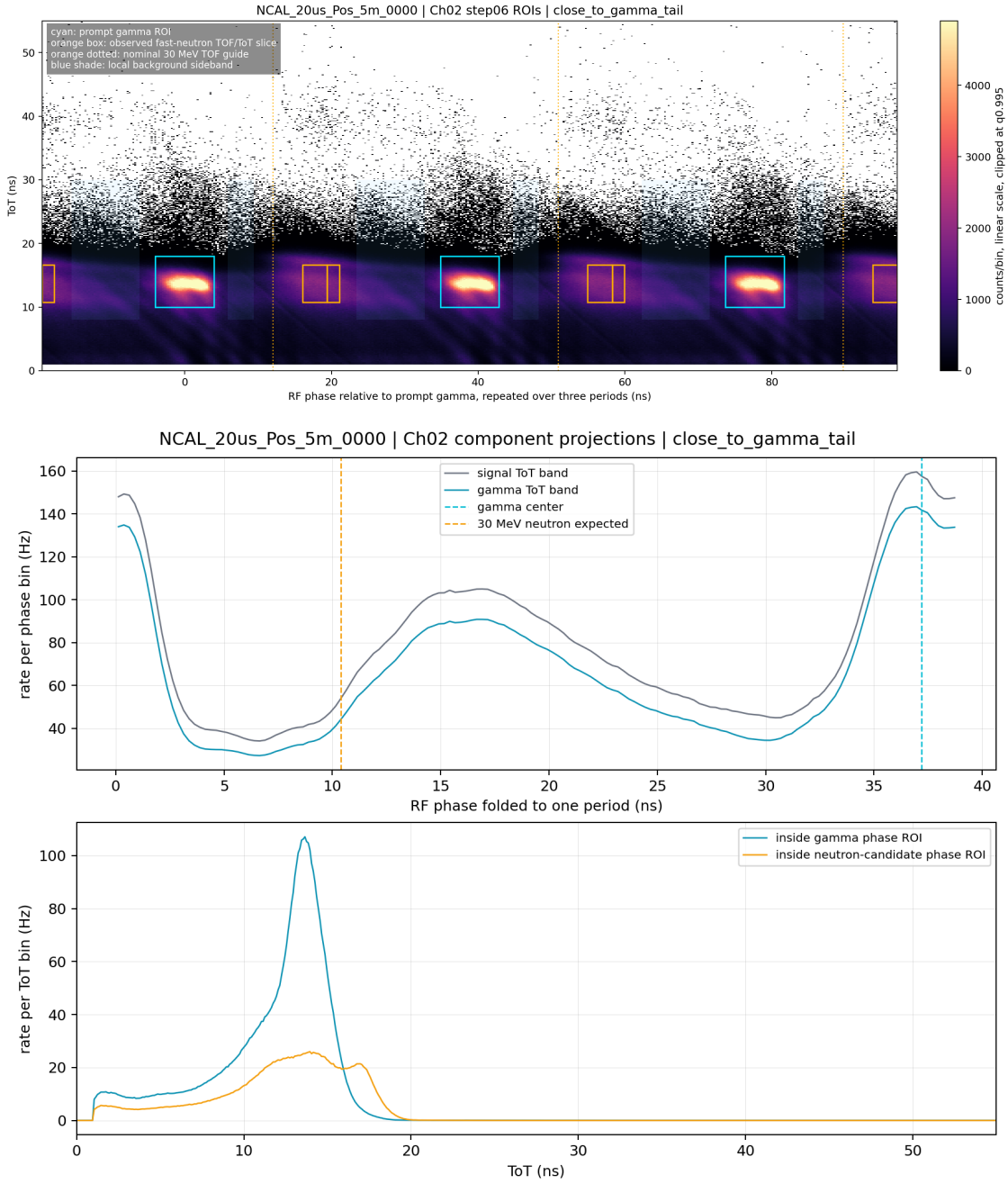


Figure 6.6: Representative step-06 gamma–neutron component diagnostics for the $5.0\,\mathrm{m}$ run. Top: the phase–ToT overlay shows the prompt-gamma ROI, the nominal 30 MeV TOF guide, the observed fast-neutron candidate slice, and the local background sidebands. Bottom: the one-dimensional phase and ToT projections show why the extracted neutron quantity is treated as a constrained fast-neutron slice rather than as the total neutron population.

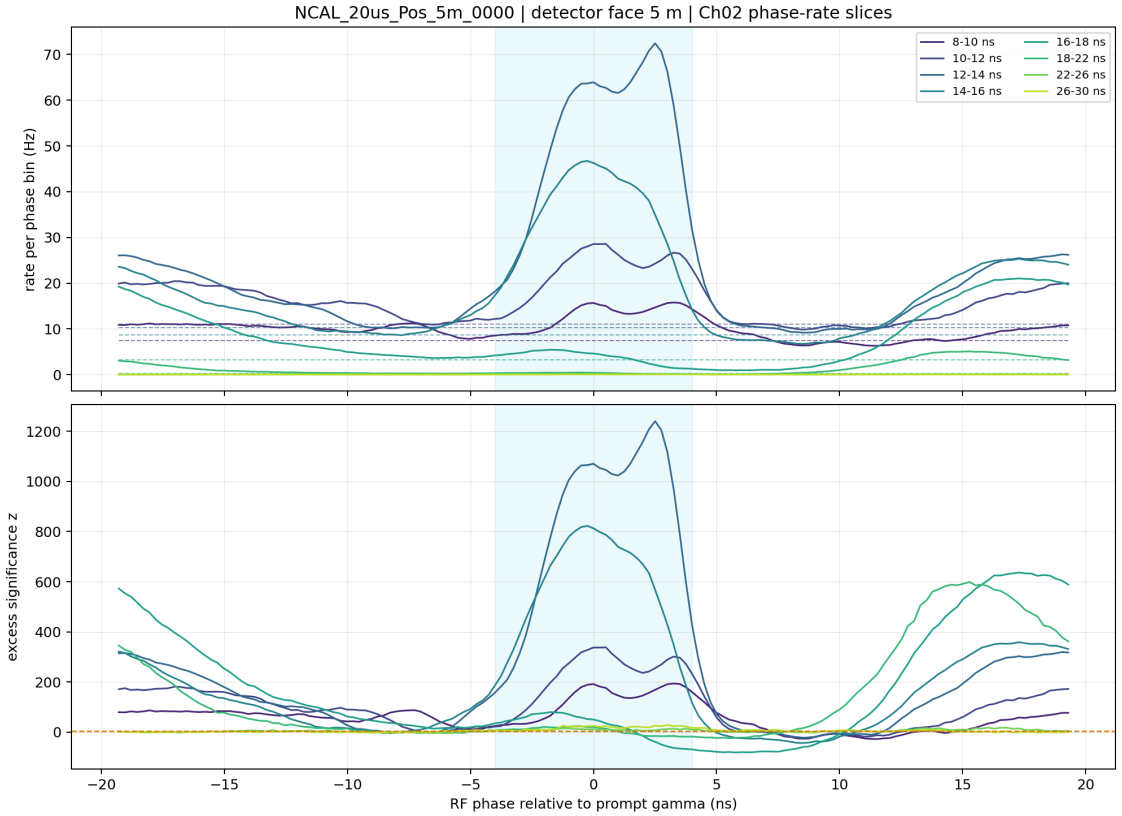


Figure 6.7: Representative step-07 one-period phase-rate diagnostics for the separated 5.0 m case. The prompt-gamma band is centered at phase zero, and the delayed excess structure is inspected slice by slice in ToT before two-dimensional signal and background regions are finalized.

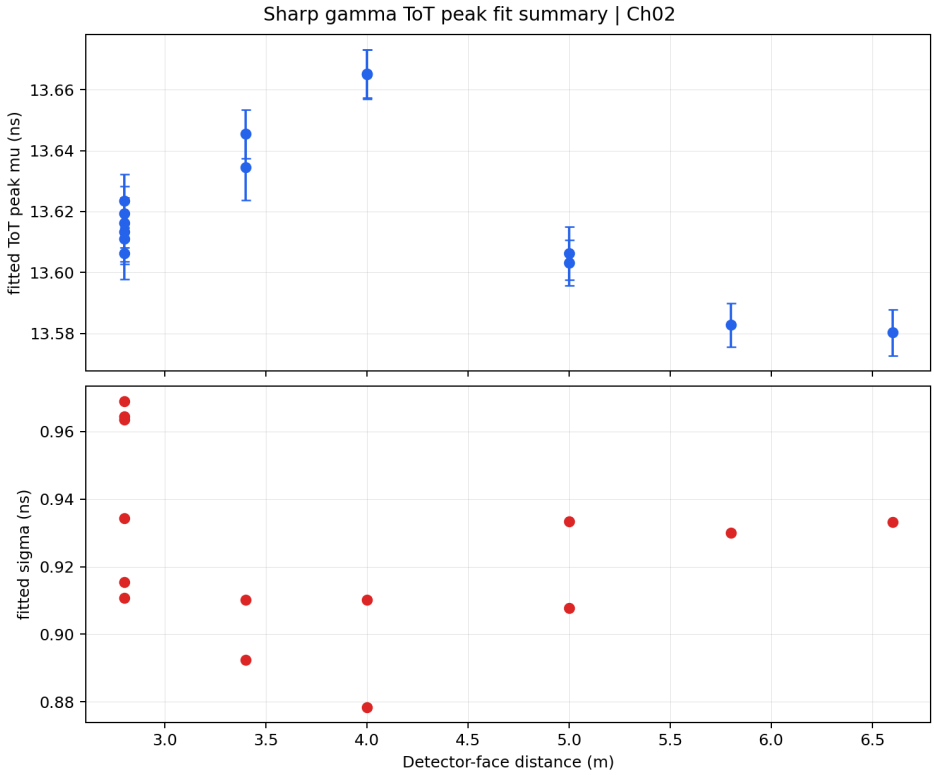


Figure 6.8: Summary of the step-08 narrow prompt-gamma ToT fits for channel 2. The fitted peak center remains stable across the completed position-scan runs, which justifies using μ_γ as the run-by-run centering proxy for the prompt-gamma gate. The quoted fit errors are statistical under the chosen local model and should not be interpreted as the full systematic uncertainty of the pulse-width distribution.

is fitted in a narrow neighborhood around the cut-band maximum. The selected prompt-gamma reference is the fitted vertex,

$$\phi_{\mathrm{ref}} = -\frac{b}{2a}, \tag{6.9}$$

with fallback to the seed maximum if the fitted curvature is not concave or if the vertex escapes the local fit window. This choice keeps the reference tied to the prompt peak while reducing one-bin jumps that can appear when the cut-band top is slightly split or asymmetric.

Two further diagnostics are then used. Step 09 varies the prompt-gamma ToT half-width around μ_γ and rebuilds the anchor to test window sensitivity. Step 10 compares the resulting local-fit reference against raw-maximum and CFD-style alternatives. In the present working document these studies are treated as systematic-control diagnostics rather than as a substitute for a full publication-level uncertainty budget. The prudent scientific statement is therefore that the current prompt-gamma anchor is designed to be sub-bin stable and auditable, while the final uncertainty budget should also absorb gate-width choice, binning dependence, and overlap-dependent background structure.

6.1.6 Current production status

In its current form, the DOGMA chain should be regarded as the robust production baseline for the commissioning data set. It already incorporates the main solutions required by the present beamtime analysis: pulse-level cleaning, explicit `ch0` auditing, consistent trigger- and channel-0-referenced observables, RF-folding diagnostics, within-run phase segmentation, first-pass background estimation, component studies for the position scan, and a multi-stage prompt-gamma anchoring strategy.

At the same time, not every diagnostic should yet be read as a finalized physics result. In particular, direct neutron numbers extracted from overlap runs remain constrained ROI measurements rather than total neutron yields, and the prompt-gamma anchor studies are best interpreted as the current methodological baseline for a later fully synchronized remote production. This distinction between validated workflow and finalized physics observable is essential if the document is to remain both reproducible and academically honest.

6.2 CAEN analysis

The CAEN-based path is presently an analysis-adjacent commissioning branch rather than the synchronized physics baseline of the beamtime. In practice, the DT5725S plus *CoMPASS* chain has been used to verify detector response, threshold behavior, rate stability, and the general readiness of the front-end electronics. Those studies are scientifically useful because they constrain whether later timing or pulse-shape observables can be trusted, but they do not yet form a curated repository-scale result chain analogous to the DOGMA cleaned-data workflow.

Accordingly, the current CAEN analysis should be described conservatively as a validated instrumentation and signal-quality study. A fuller CAEN analysis section would eventually need curated list-mode or waveform datasets, explicit calibration logic, dead-time treatment, and figure-level outputs that can be cross-compared with the DOGMA observables on equal footing.

6.3 Oscilloscope-supported analysis

Oscilloscope measurements are not the primary archival data source of the present campaign, but they remain essential as a support-analysis instrument. Direct waveform inspection is the fastest way to identify baseline motion, overshoot, reflections, saturation, polarity mistakes, PMT light leaks, or cable problems before those effects appear indirectly in digitized timing or ToT observables. In that sense, oscilloscope work belongs to the analysis chapter because it constrains the physical interpretation of later digital observables even when the oscilloscope traces themselves are not the final published dataset.

At the present stage, these oscilloscope-supported studies are mainly qualitative or semi-quantitative diagnostics. They define the boundary conditions under which the digitized data can be trusted, but they do not yet constitute a stand-alone repository-scale analysis branch with synchronized result figures comparable to the DOGMA chain.

6.4 DiRICH analysis context

The DiRICH-based readout path remains scientifically relevant as a comparison branch because earlier NCAL work used that chain and because comparative readout studies are important for detector-development decisions. The key comparison questions are not only hardware oriented. They also concern timing-reference stability, data integrity, pulse-width behavior, analysis overhead, and the extent to which later particle-identification observables can be extracted reproducibly.

At present, however, the synchronized analysis products curated in this repository are DOGMA based. The DiRICH section is therefore intentionally brief and should be read as methodological context rather than as a fully developed results section. A mature DiRICH analysis treatment would require curated aligned datasets and direct observable-level comparisons against the DOGMA baseline.

Chapter 7

Commissioning and First Validation Workflow

7.1 Purpose of the next work package

Once stable communication with the DT5725S has been established, the next work package is no longer device discovery but controlled validation. The practical objective is to produce a small set of well-documented reference runs that can be compared later against detector changes, threshold adjustments, firmware changes, or analysis updates. In other words, the transition from "the board can be opened" to "the readout chain is scientifically usable" requires a reproducible workflow rather than an isolated successful test.

For the present campaign, this next step is especially important because the DAQ chain is expected to support later detector-comparison work and eventual neutron-beam measurements. A reference validation dataset should therefore be treated as an intermediate deliverable in its own right. It anchors later decisions on channel mapping, operating thresholds, pulse polarity, firmware choice, and file-format handling.

7.2 Recommended validation sequence

The initial validation sequence should remain deliberately simple. First, a short pedestal or no-source run should be recorded to document the electronic baseline, pickup environment, and spontaneous trigger behavior. Second, a single detector channel should be connected and validated under the same DAQ settings used during board discovery, so that waveform stability and trigger-rate response can be checked without the ambiguity introduced by multi-channel operation. Third, only after this single-channel behavior is stable should threshold adjustments or additional detector channels be introduced.

For the current DT5725S configuration with DPP-PSD firmware and *CoMPASS*, the minimum acceptance test for a reference run is straightforward: the board must be rediscovered after a fresh start, a selected channel must produce a stable rate and visible pulses, and the saved data file must be consistent with the configuration used during acquisition. If any of these conditions fails, the run should be classified as a commissioning test rather than as a physics-quality reference run.

Where detector conditions permit it, a very small threshold scan is already useful at this stage. Even a coarse set of operating points can reveal whether the observed rate is dominated by electronic noise, low-amplitude background, or clearly resolved detector pulses. This is more

informative than relying on a single visually satisfactory setting, and it provides a first quantitative handle on the stability margin of the chosen operating point.

7.3 Run record and promotion criteria

Each accepted validation run should preserve more than the raw event file alone. At minimum, the run record should include the saved *CoMPASS* configuration, the board serial number, the firmware identifiers reported by the CAEN libraries, the driver version or package used on the acquisition PC, the active channel list, the detector connected to each channel, the applied threshold and polarity settings, and a short operator note stating the source or pulser condition. This information is lightweight to archive, but it is essential if later discrepancies between runs are to be interpreted correctly.

Promotion from validation mode to routine data taking should require three conditions. First, board communication must remain stable across repeated restarts of both the software and the device. Second, the observed count rate and displayed pulses must remain consistent over a run long enough to exclude a transient startup effect. Third, the complete acquisition state must be reproducible by another operator using only the stored configuration and laboratory notes. If these conditions are met, subsequent measurements can be interpreted as detector studies rather than as continued infrastructure debugging.

This work package also defines the point at which a future branch in the DAQ strategy becomes meaningful. If the reference runs show that the DPP-PSD configuration already provides the observables needed for detector discrimination and timing studies, then the present firmware can be retained for the next stage. If, however, the later analysis requires full trace-level treatment of baselines, charge integration windows, or pulse-shape development, then the justified next step is a planned transition to the x725S waveform-recording firmware together with a waveform-oriented acquisition program.

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